



GE VERNOVA

ADMS

16.25.1

File Updater Installation Guide

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Change History

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Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Mar 25, 2024	ADMS 16.7.0: • Updated document format and styling.
Jul 3, 2024	ADMS 16.10.0: • Updated Generating Identity Keys.
Sep 26, 2024	ADMS 16.12.0: • Adding digital signature to files synchronization process.

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1. Introduction

The File Updater server and the File Updater client are standalone applications that are part of the ADMS product suite.

The File Updater server and the File Updater client provide automatic synchronization between the defined sets of folders of the model data server and the local folders of ADMS applications.

The File Updater server and the File Updater client are configured and managed using the File Updater Configuration Editor.

1.1. File Updater Server

The File Updater server provides the following core functions:

- Monitors sets of folders for new, modified, and deleted files
- Publishes changed files to the FTP/SFTP server or to a local folder when in the Local mode
- Synchronizes file deletion
- Leverages private key authorization (for SFTP connection)
- Performs atomic file swap-outs (rename, not overwrite)
- Supports compressed file transfers
- Supports digitally signed file transfers
- Provides detailed logged messages for troubleshooting and other information
- Supports Microsoft Windows 64-bit platforms
- Can run as a service
- Supports non-ASCII (internationalized) file names

1.2. File Updater Client

The File Updater client provides the following core functions:

- Synchronizes sets of folders with an FTP/SFTP server
- Synchronizes file deletion
- Leverages private key authorization (for SFTP connection)
- Performs atomic file swap-outs (copy-rename, not overwrite)
- Supports compressed file transfers
- Supports verification of digital signatures embedded in files
- Provides detailed logged messages for troubleshooting and other information
- Offers an Update Now capability

- Supports Microsoft Windows 64-bit platforms
- Can run as a service
- Supports non-ASCII (internationalized) file names

1.3. Background

The ADMS servers and clients have full copies of electrical models with a model granularity on a substation basis. This approach facilitates online changes. For instance, when a new switch is added to a particular feeder, updating the model used by the real-time system only consists of loading the updated model of the substation of interest.

The ADMS File Updater software is responsible for keeping the ADMS servers and clients up to date in terms of data files. In addition to the station model files (for example, `substationname.mod`), there are XML-based toolbar, configuration, and tabular display files, and bitmap files that also need to be updated as changes arise.

The following image displays the File Updater flowchart for the SFTP connection.

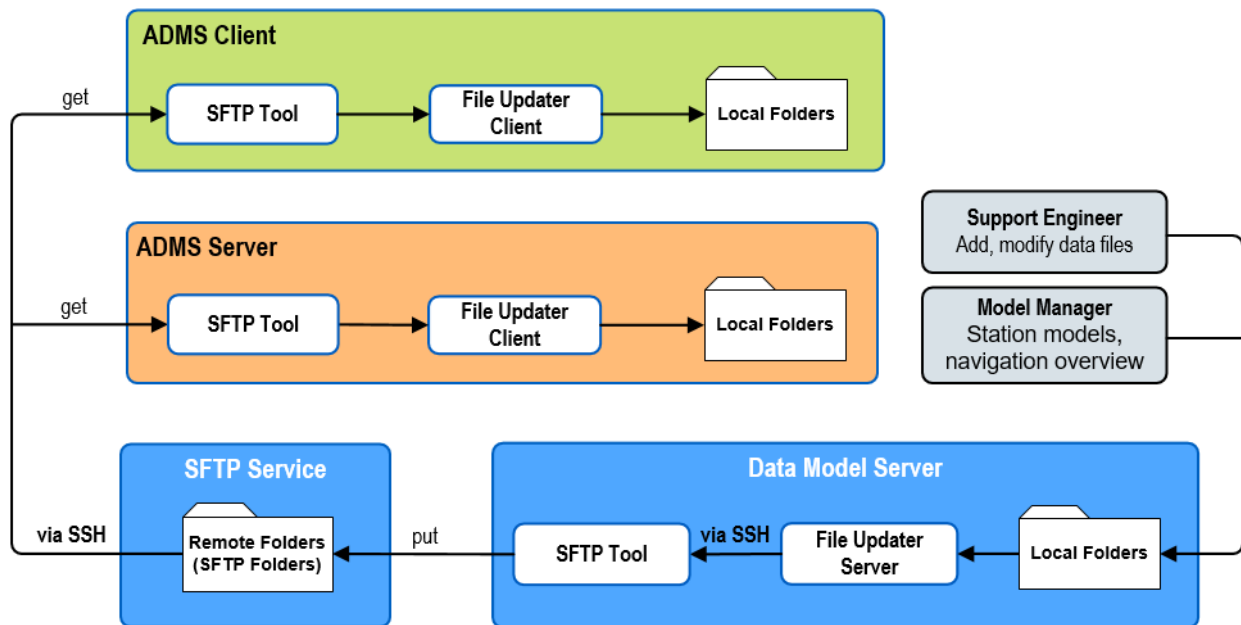


Figure 1. SFTP File Updater Component Diagram

1.3.1. File Updater Server Workflow

1. During startup, if Digital Signature is enabled, File Updater Client will check that the digital certificate is installed and start date and expiration dates are valid. If certificate is missing or invalid, the process will stop immediately.
2. After the certificate check, the File Updater server locates the UpdatedFilesLog file defined in the File Updater server configuration file. The UpdatedFilesLog file contains a list of previously

synchronized local files that were sent to a remote server. If the File Updater server fails to locate the UpdatedFilesLog file, it creates a new one.

3. Every folder specified in the File Updater server configuration file is synchronized at periodic intervals. Those intervals are defined via the Time Delay field in the Directory Entry panel of the File Updater Configuration Editor (see [Using Configuration Editor to Define Synchronized Directories](#)).
4. The File Updater server initiates timers with the Delay parameter values and synchronizes directories specified in the Directory Entry list. A single folder is processed in the following way:
5. The File Updater server waits for directory timers to trigger.
6. When a directory timer (Time Delay) triggers, the File Updater server examines the folder for new or modified files by calculating the Cyclic Redundancy Check (CRC) value, checking the size, and reading the creation date of every file.
7. The File Updater server compares values with those specified in UpdatedFilesLog. If differences are found, the file is treated as new or modified.
8. While the file is prepared to be copied to the remote folder and if the option "Use Digital Signature" is selected, each file is digitally signed using the digital certificate previously installed in the machine. This certificate is identified by the "Certificate Name" and retrieved from the certificate store identified by the "Certificate Store Name" from the configuration. It is highly recommended to install these certificates in "root", also known as Trusted Root Certification Authority. The certificate name is usually displayed in the first column in the list of certificates displayed by certmgr.exe. While digitally signing files, File Updater may stop if the certificate has expired.
9. When the list of modified files is complete, the files are sent via FTP/SFTP/File Updater "put" methods to the corresponding remote folder specified in the configuration file.
10. To prevent name collisions, the new and modified local files are renamed to temporary names (FileName + "_tmp" suffix) and sent to a remote folder. The existing modified files in the remote folder (old files to be replaced) are also renamed to temporary names (FileName + "_tmp2" suffix). Then the new files are renamed to the original names, and the old files are removed.
11. After all files from the local directory are sent to the remote folder, the File Updater server starts processing the next local folder. If an error occurs while sending a file, two options are available:
 - If the "Continue On Error" option is selected in the Directory Entry panel, the File Updater server performs "one-step" upload of the files "as is" by directly overwriting files with the same names on the remote server. If an error exists, the application skips the failed file and continues sending the list.
 - If the "Continue On Error" option is not selected, the File Updater server aborts synchronization of the current folder and starts processing the next folder.
12. After all folders have been synchronized, the File Updater server places the UpdatedFilesLog file to a specified remote folder. At that moment, the UpdatedFilesLog file contains a list of all successfully synchronized files.

The following image shows the File Updater server synchronization workflow.

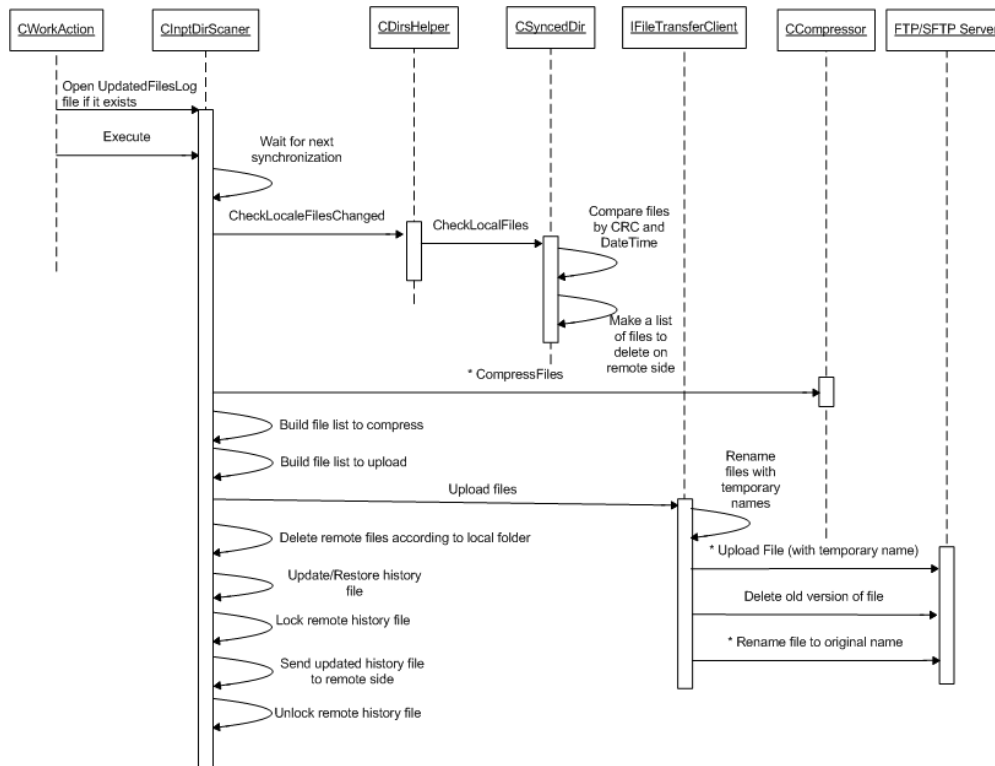


Figure 2. File Updater Server Synchronization Workflow

1.3.2. File Updater Client Workflow

1. During startup, if Digital Signature is enabled, File Update Client will check that the digital certificate is installed and start date and expiration dates are valid. If certificate is missing or invalid, File Updater Client will stop immediately.
2. When the timer defined in the Directory Entry panel of the File Updater Configuration Editor is triggered, the File Updater client starts the synchronization process.
3. The File Updater attempts to download the UpdatedFilesLog file. Meanwhile, the UpdatedFilesLog file can be in use by the File Updater server for updates. Collisions are prevented by the Lock File process, where, during the UpdatedFilesLog catalog update, the server creates a lock file using the UpdatedFilesLog's name with the word "LOCK". Once the update is complete, the lock files are deleted.
4. Before the client attempts to download the UpdatedFilesLog file, it looks for the lock file. If the lock file is found, the client waits until it is deleted, and then continues synchronization.
5. The client downloads the UpdatedFilesLog and checks for differences between the files listed in the UpdatedFilesLog and the files in its local and temporary folders. The temporary folder can contain newer versions of files than the local folder. This occurs when a previous download process was not successful, and some files remained in the temporary folder.
6. Once the list of modified files is ready, the client downloads these files from remote folders to the temporary folder. Downloaded files have temporary names (FileName + "_tmp" suffix).

7. If the download was successfully completed, the digital signature within each file is verified (if configured) using the same certificate the files were signed with, the files are uncompressed and verified by the CRC check. The CRC check ensures that a remote set of new files is not corrupted (files could be deleted, changed, or overwritten by third-party applications before the client attempted to download them). If the CRC check is passed, the old files are deleted, and newly downloaded files are copied to the target local folder and renamed from temporary to original file names.
8. As an additional layer of security, after the CRC is checked, the digital signature is validated using the certificate information embedded in the signature. This requires that the right certificate is installed in the local machine for proper signature decryption. If files are digitally signed by server but same functionality is disabled in the clients, the latter will try to copy the files without removing the digital signature which will result in CRC mismatch errors and files will not be copied. To fix this, both server and client instances should enable or disabled digital signature option accordingly.
9. If an error occurs during the download process, two options are available:
 - If the "Continue On Error" option is selected in the Directory Entry panel, the client ignores these errors and continues to download files.
 - If the "Continue On Error" option is not selected, the client aborts the download process. All files that were successfully downloaded to a temporary folder are deleted.

These actions are performed periodically as specified in the configuration file.

The following image shows the File Updater client synchronization workflow.

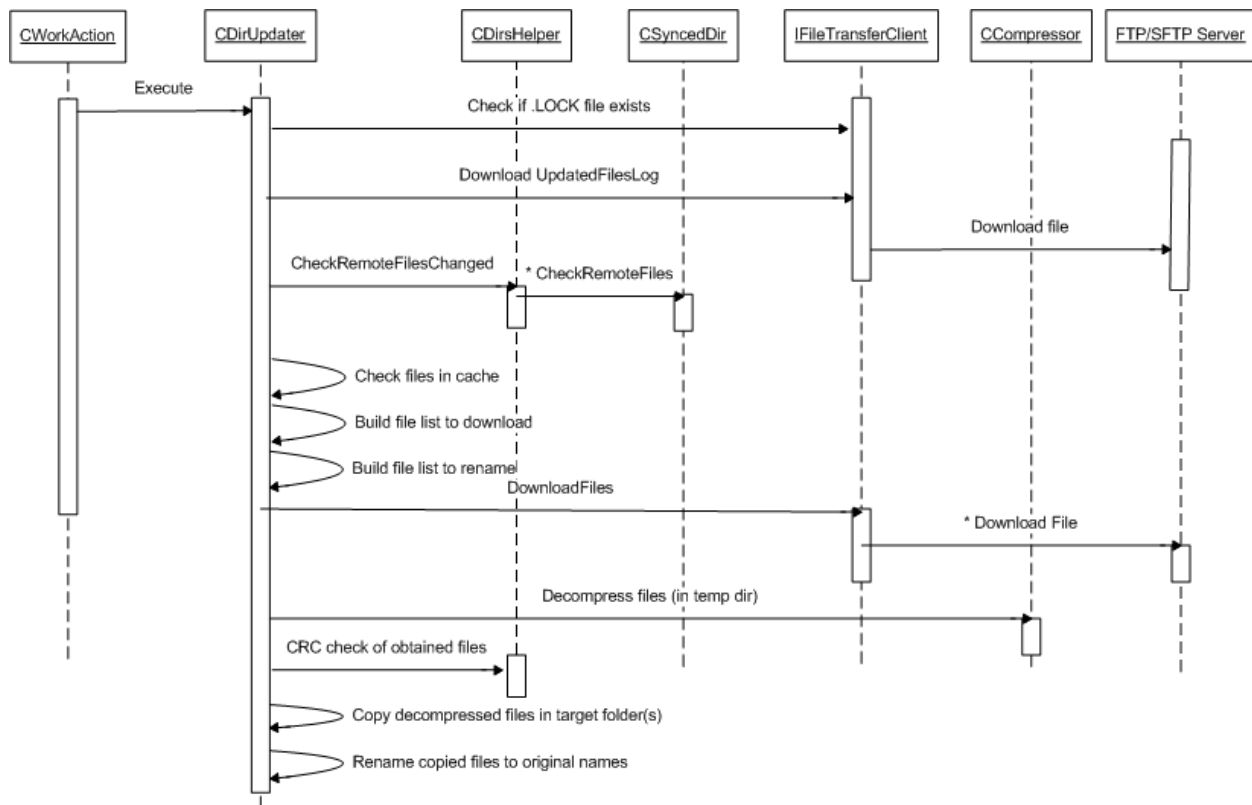


Figure 3. File Updater Client Synchronization Workflow

1.4. Restrictions

Configuration files must contain only English language characters. Unicode is currently not supported.

1.5. Terminology

This section defines terminology used in this document to describe the ADMS File Updater functionality:

- **ADMS Servers:** The main ADMS server applications, including the DMS server (NetSrv), OMS server (NetServer), DNAF server (DnafSrv), etc. The servers listen for ADMS client connections, keep them up to date with dynamic data, and provides data query results.
- **ADMS Clients:** The main ADMS client components. A client runs as a plug-in within Browser. A client includes the geographic view, tabular displays, area world view, pop-up dialog boxes, and navigation within Browser.
- **ADMS File Updater server:** This application (service) runs on a server computer, monitoring the data files that need to be published and updating the manifest files.
- **ADMS File Updater client:** This application (service) runs on the ADMS client and server

computers to keep them up to date with the latest data files.

- **File Updater "application(s)":** This term can mean either the File Updater server or the File Updater client. This term is used when describing features that are common to both the File Updater server and the File Updater client.
- **Local folder:** A file directory on the same host as the File Updater application.
- **Remote folder:** A file directory on the FTP/SFTP server or on a local host when in the Local mode.
- **Account:** This specifies the logon account for the File Updater application(s) when running as a Windows service.
- **Updated files log:** An XML-based file that describes the state of the remote folders (lists all files and their sizes. CRC is used as a checksum to detect accidental alteration of data. This file is created by the File Updater server, and it is used by the File Updater client to determine which local data files should be updated.
- **Configuration file:** An XML-based file that contains settings (configurations) for the corresponding File Updater application and that has a specific structure. An empty configuration file can be generated by the File Updater application.
- **Atomic swap-out:** A binary operation that consists of copying a large file from one computer to another with a different file name instead of replacing the existing file on the other computer. Once the copying process is complete and the file is validated, the original file is deleted and the copied file is renamed correspondingly. For example, the atomic swap-out of the Raven.mod file consists of the following operations:
- **Digital Certificate:** A file that contains information (cryptography keys) to encrypt and decrypt digital information. This file was created following the X. 509 public key infrastructure (PKI) standard. This certificate shall be installed in the local machine before performing digital signature operations. A certificate can be identified with a certificate name.
- **Certificate Store:** A container of certificates installed in a Windows computer that can be accessed through the Certificate Management Tool. A certificate store can be identified via a certificate store name, e.g., "root" is the name of the Trusted Root Certification Authorities certificate store.
- **Digital Signature:** A digital signature is an electronic, encrypted, stamp of authentication on digital information such as email messages, macros, or electronic documents. A signature confirms that the information originated from the signer and has not been altered.
 - Copy Raven.mod to another computer and name the copied file Raven_tmp.mod.
 - Validate the Raven_tmp.mod file.
 - Delete the existing Raven.mod file on the computer that is currently in use.
 - Rename Raven_tmp.mod to Raven.mod.

This process eliminates the time period when a file is transferring.

File Updater uses the atomicity of operations to ensure consistency of data that is updated. If File Updater fails to update one or more files failed during a synchronization session, the whole

synchronization attempt is considered as failed, and the whole update operation is canceled. To support partial updates, the "Continue On Error" option must be selected (see [Using Configuration Editor to Define Synchronized Directories](#)).

In addition, to ensure a successful synchronization of a batch of interdependent folders, the folders can be defined as "dependent". In this case, if a synchronization failure occurs for one folder in the batch, the synchronization for the others is also canceled (see [Defining Dependent Folders](#)).

2. Setting up the FTP and SFTP Connection

The File Updater components provide end users with the choice of using FTP or Secured FTP (SFTP) in passive mode. File Updater uses Windows FTP and SecureFX, both of which support domain accounts.

Note

File Updater does not support FTP/SFTP in active mode.

Because File Updater uses external tools to transfer files from remote servers, specific access issues (for example, support of domain user names to get access to the FTP/SFTP servers) should be considered, taking into account the actual tools used.

File Updater has no functionality to manage permissions. Permissions should be properly configured on the FTP/SFTP server that verifies user credentials and decides whether a specific user is allowed to access the server's resources.

This document only gives a sample description of the FTP/SFTP configuration. There are many different ways to configure FTP/SFTP permissions, but this issue is beyond the scope of the File Updater documentation, as configuring these permissions is a system administration issue.

The following prerequisites apply to all supported and certified configurations of the File Updater server and File Updater client. For a list of the platforms, see the "Supported Configurations" section in the *ADMS Release Notes*.

To make sure you have the latest information, including whether each prerequisite is for the File Updater server, File Updater client, or both, review the "Certified Third-Party Products and Packages" section in the *ADMS Release Notes*.

Set the SSH connection according to the *Secure Shell (SSH) User Guide*.

Note

The File Updater server and client applications can be installed and run on the same computer.

2.1. SFTP Installation and Configuration

To use the File Updater via SFTP in passive mode, install the VanDyke VShell server and the VanDyke SecureFX SSH client applications on the File Updater server computer (a host that is used for file storage). On the File Updater client computer, install the VanDyke SecureFX SSH client application.

When the VanDyke applications are installed, you need to create identity keys (public key and private key), define SFTP folders, and configure the authentication settings.

Note

VanDyke VShell version 3.8 or later and VanDyke SecureFX SSH version 6.7 or later are

recommended. When working with earlier versions of SecureFX, the following problems were discovered:

- SecureFX 6.1.0–6.1.3 have minor problems with files that have the letter "s" as the second letter in the file name.
- SecureFX 4.3 has problems with Unicode file names.
- SecureFX 2.5 does not support identity key command-line authorization.
- SecureFX 6.7 and earlier do not support the semi-colon (";") in remote file names.

2.1.1. SSH Connection Setup

The SSH connection must be set up according to the *Secure Shell (SSH) User Guide*.

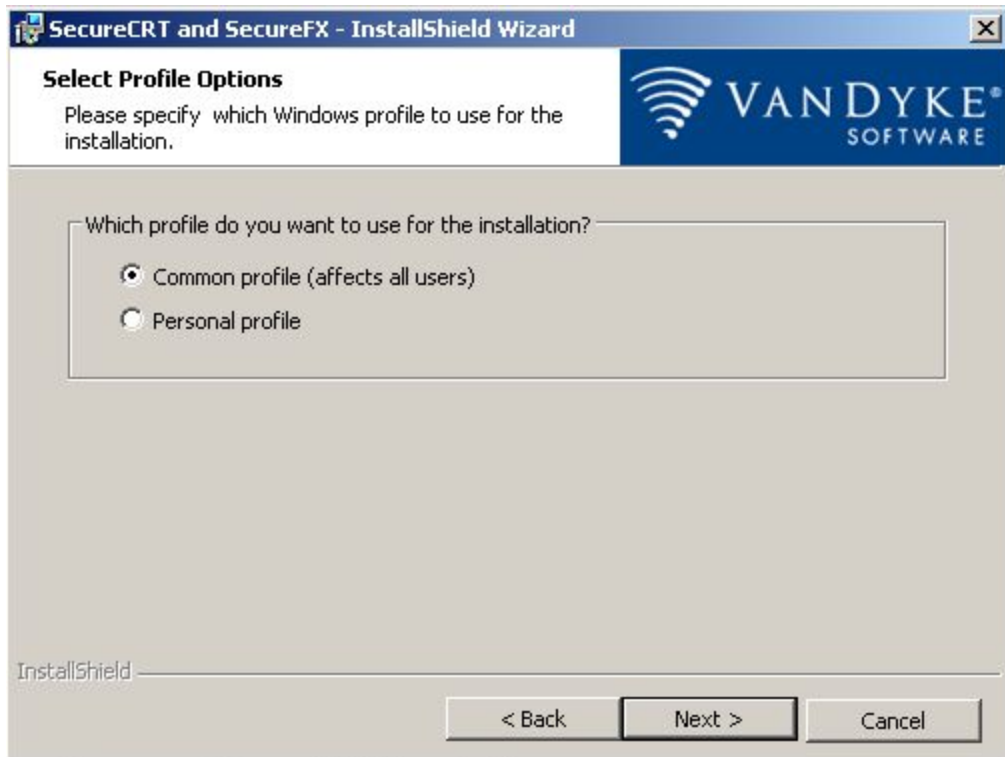
Requirements for the SSH connection setup are as follows:

- Identity keys must be generated without a passphrase (passphrase blanked).
- Users with credentials that are used by the File Updater application should have rights to access the SFTP server (SSH2 over SFTP permission).
- Users with credentials that are used by the File Updater server should have write and create permissions to folders that are specified in the File Updater server configuration file.
- Users with credentials that are used by the File Updater client should have read permission to folders that are specified in the File Updater client configuration file.

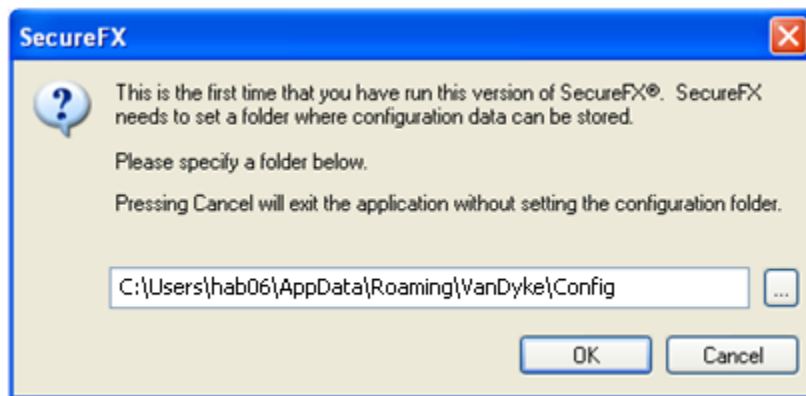
It is recommended that you set private key authorization to be used first (currently, the File Updater application only uses the private key without passphrase authorization).

2.1.2. Installing VanDyke SecureFX

1. Launch the VanDyke SecureFX installation file.
2. During the SecureFX installation, select the Common Profile option.



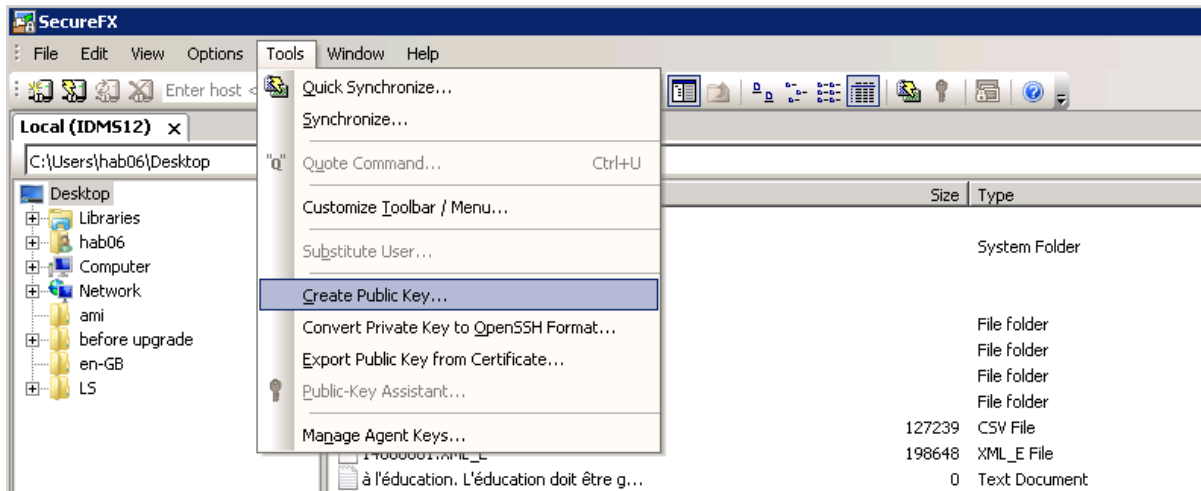
3. Start SecureFX.
4. When starting SecureFX for the first time, specify a folder where configuration data can be stored, and click OK.



It is recommended that you use the default path (the File Updater installation kit uses the default path while configuring File Updater).

2.1.3. Generating Identity Keys

1. Start the VanDyke SecureFX application and click Tools > Create Public Key.



The Key Generation Wizard appears.

2. Select the DSA key type.
3. Leave the Passphrase and Comments fields blank.
4. For Key Length in Bits, use 1024.

Note

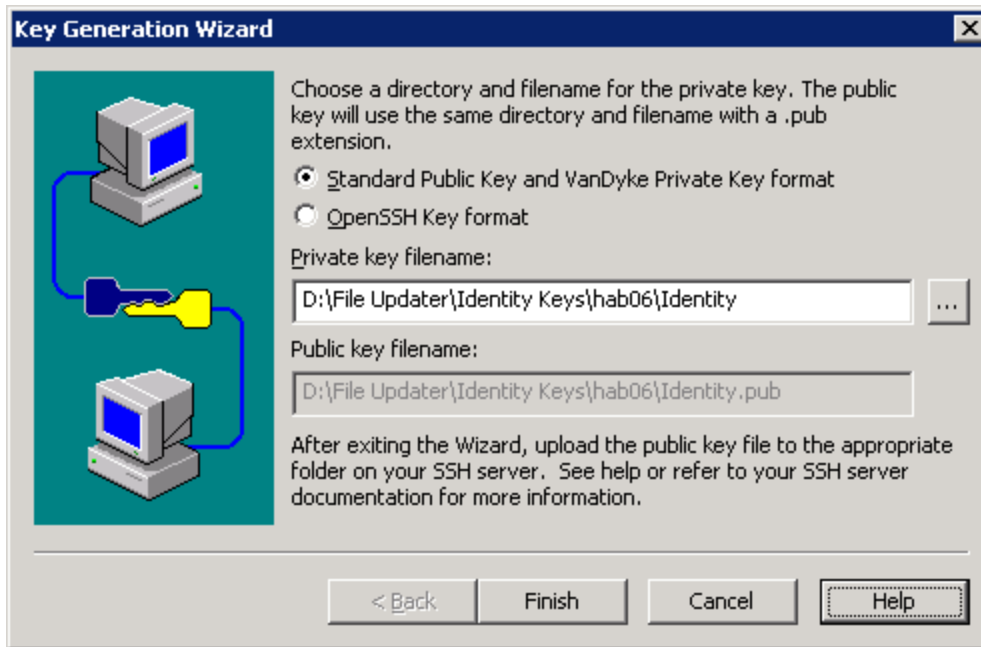
DSA keys are restricted to 1024 bits. For larger keys, select RSA, which supports 2048, 3072, and beyond.

5. Select the Standard Public Key and VanDyke Private Key Format option.

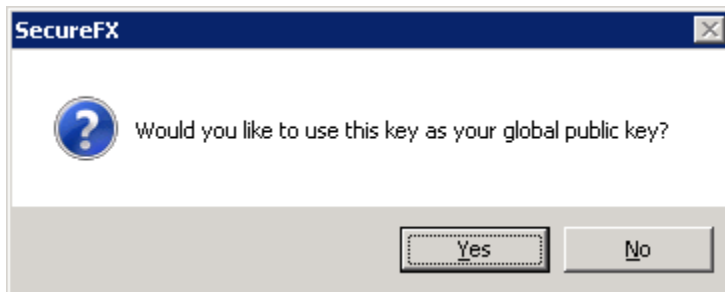
Note

When using SecureFX version 8 or later, select the OpenSSH Key Format (Legacy) option. Otherwise, File Updater errors may appear.

6. Choose the directory and filename for the private key, and click OK to save the changes.



7. Click Finish to close the Key Generation wizard.
8. Use the generated key as a global key.



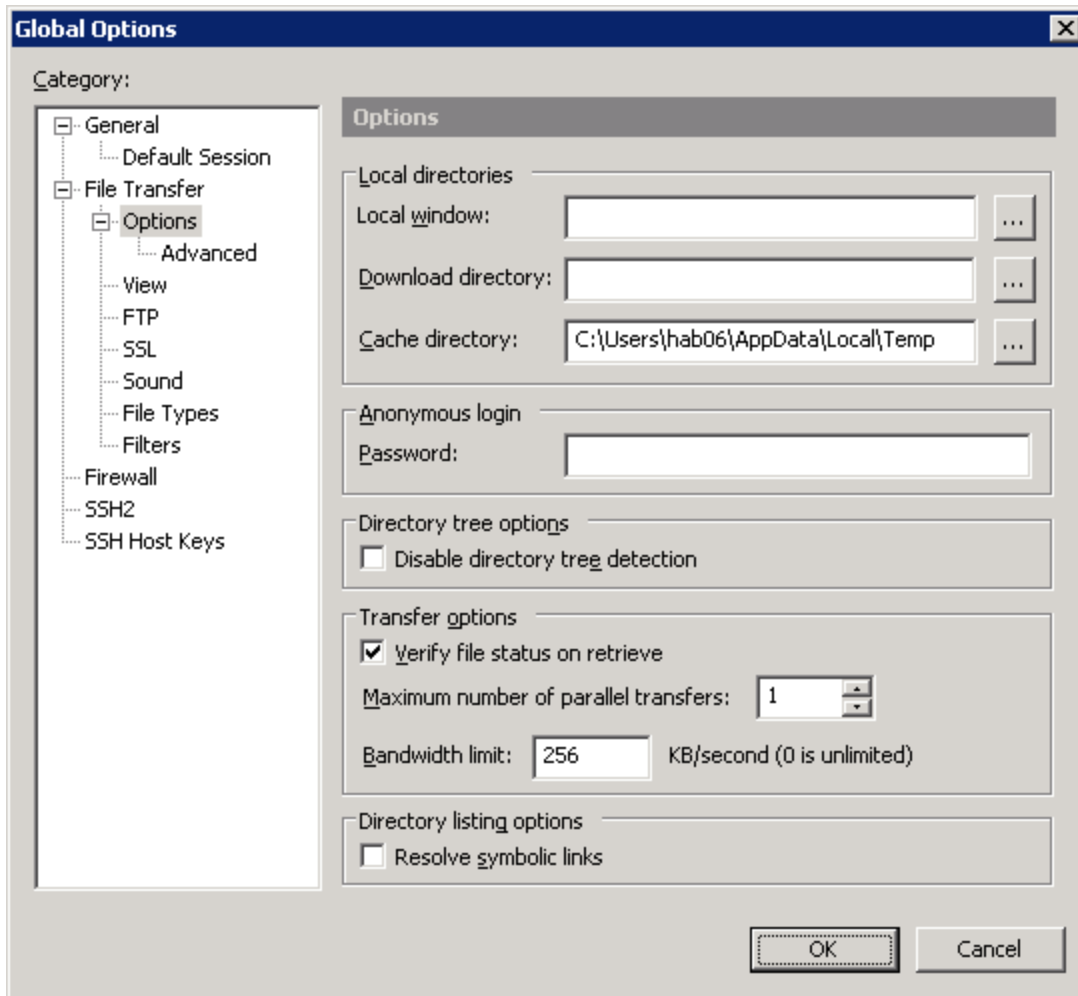
2.1.4. Configuring the Bandwidth Limit

1. In VanDyke SecureFX, click Options > Global Options.

The Global Options dialog box appears.

2. Click Options.
3. In the Transfer Options group, set the maximum number of parallel transfers and the bandwidth limit.

The transfer options allow limiting the traffic that is used by the SFTP to ensure normal functioning of the other processes. Typically, it is recommended to set the bandwidth limit to 10% of the available network bandwidth.



4. Click OK.

2.1.5. Installing VanDyke VShell

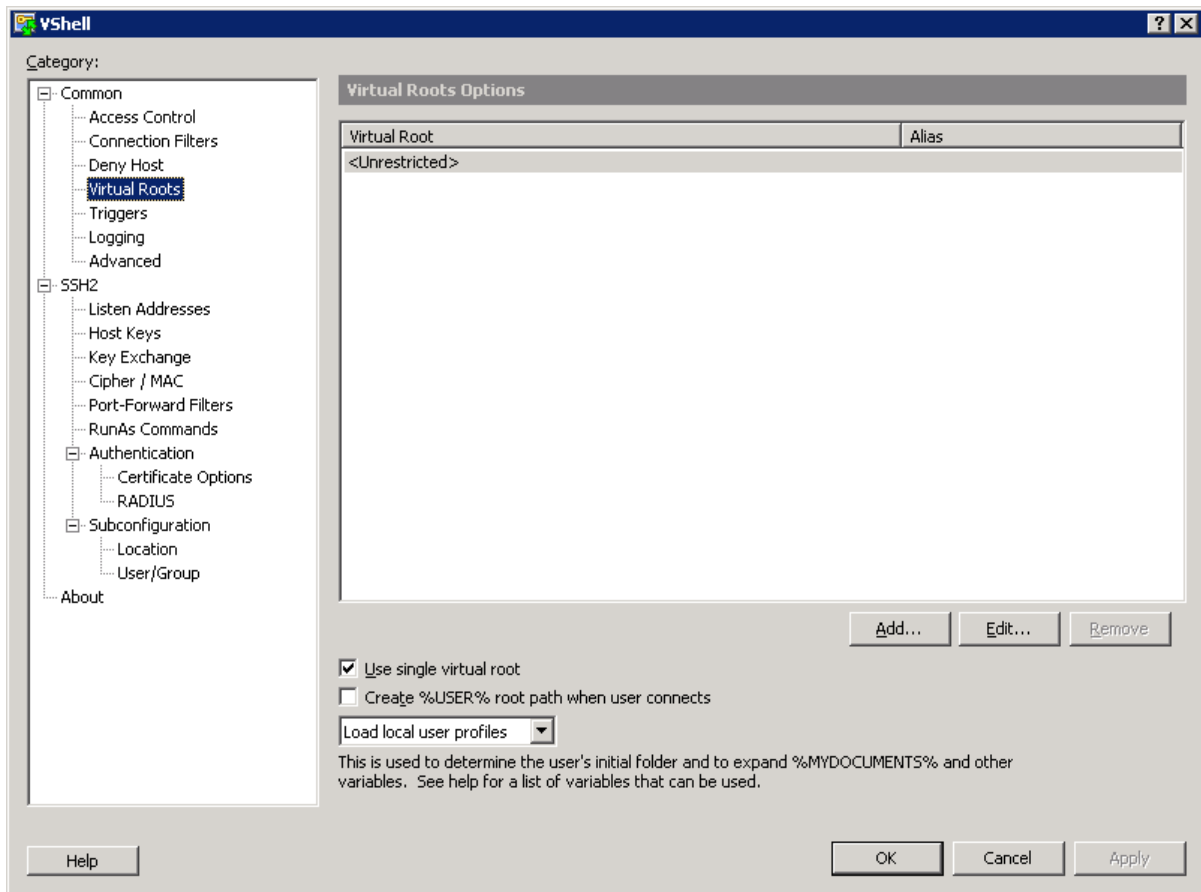
To Install the VanDyke VShell application, run the VShell setup file. Using the default VShell installation settings is recommended.

When the installation is complete, you need to configure the virtual roots and authentication properties, as described in the following sections.

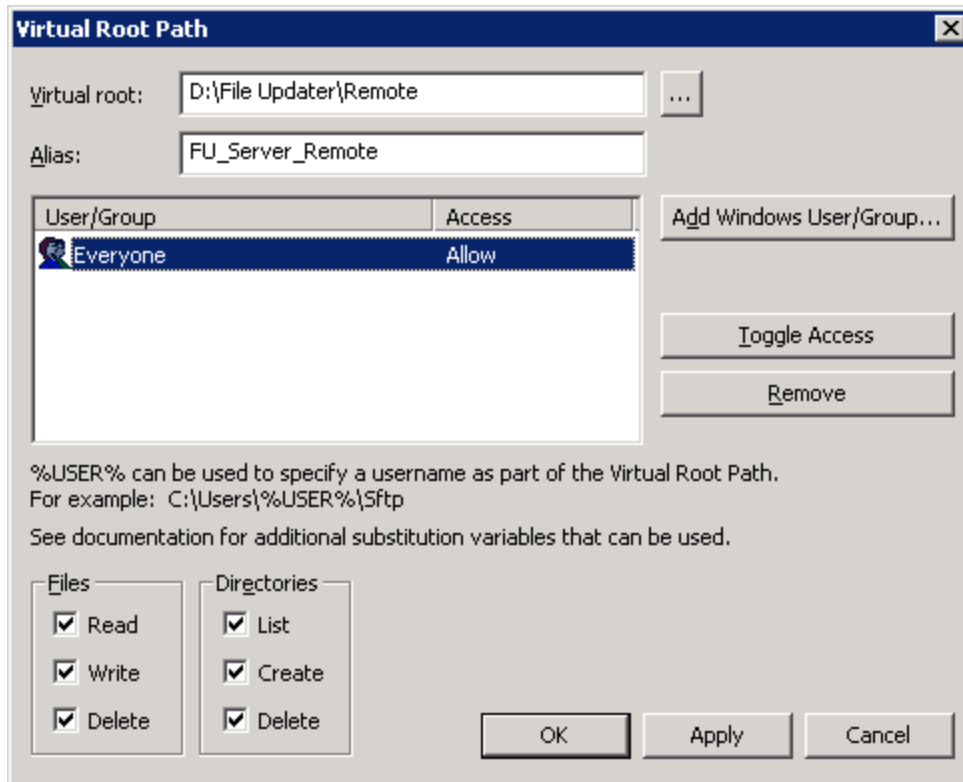
2.1.6. Configuring the VShell Virtual Roots (Defining SFTP Folders)

1. Start the VShell application.
2. Under Category, click Virtual Roots.

The Virtual Roots Options page is displayed.



3. In the Virtual Root list, select <Unrestricted> and click Edit.
4. On the Virtual Root Path form, remove the Everyone user and click OK to save the changes.
5. On the Virtual Roots Options page, click Add.
6. On the Virtual Root Path form, enter the parameters as follows:
 - Virtual Root: Set the path to the SFTP server Remote Folder as necessary.
 - Alias: Set the alias as appropriate.
 - User/Group: Add the Everyone user, and grant it permissions to work with files and directories.



Note

File Updater is case sensitive.

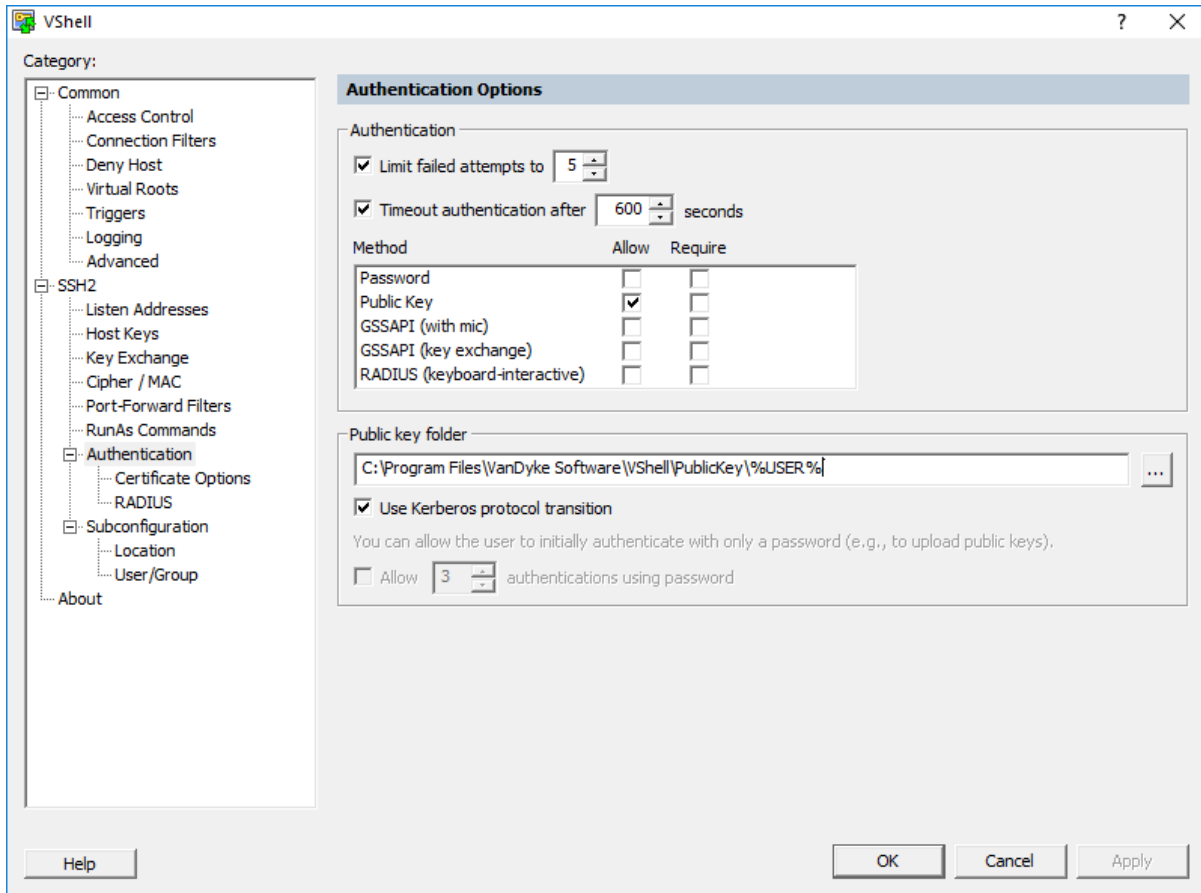
7. Click OK to save the changes and close the Virtual Root Path form.
8. Click Apply to save the changes.
9. Repeat this procedure to add more SFTP folders.

2.1.7. Configuring the VShell Authentication

VShell uses the public key to authenticate the private key supplied by the File Updater (via SecureFX) to determine whether to allow connections.

1. Start the VShell application.
2. Under Category, click Authentication.

The Authentication Options page is displayed.



3. In a folder for public keys (by default, C:\Program Files\VanDyke Software\VShell\PublicKey\), create subfolders with SFTP client user names (for example, hab06 for the File Updater client and hab07 for the File Updater server).

Note

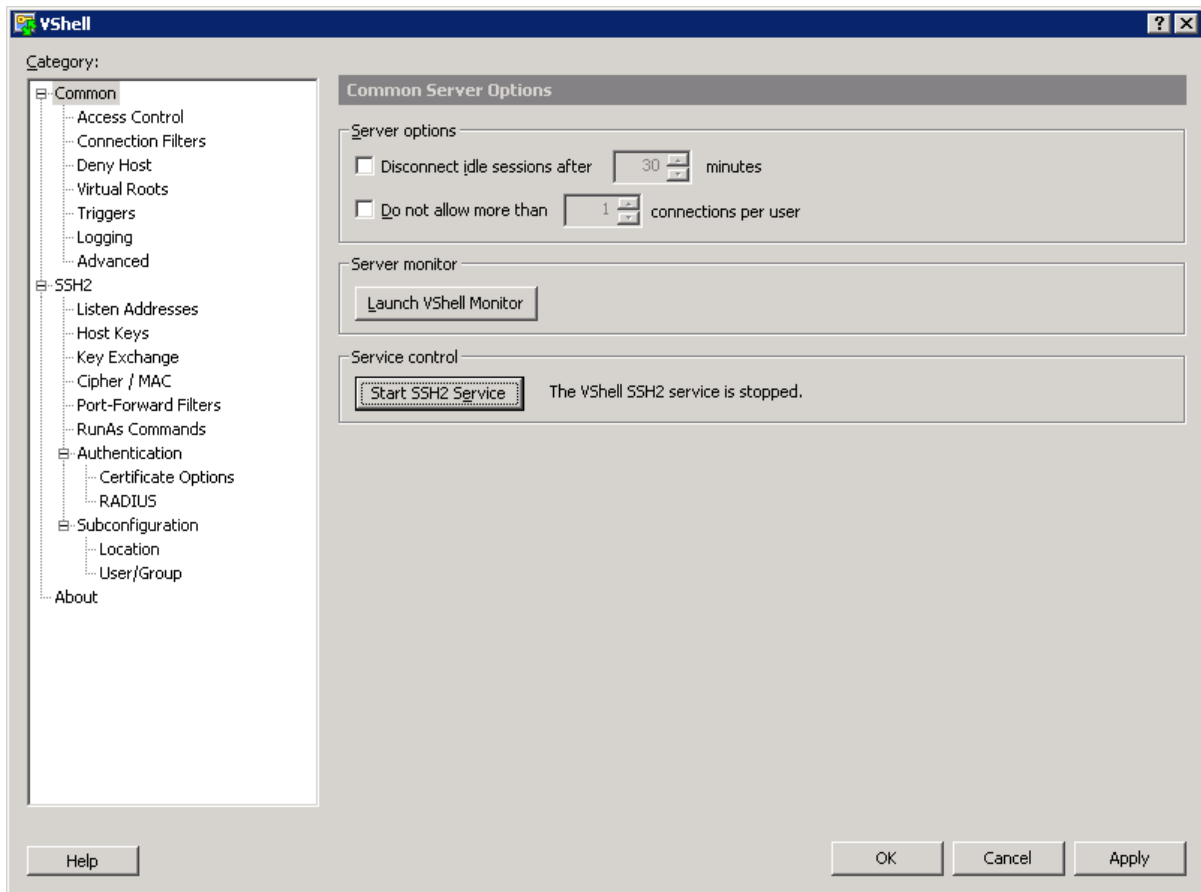
The public key folder path format requires the public key to be placed in the folder with the SFTP user's name.

4. Copy public keys generated for the File Updater client and File Updater server machines (see [Generating Identity Keys](#)) to the user subfolders in the public key folder.
5. Set the Public Key Folder path to <Path to the folder for public keys>\%USER% (for example, C:\Program Files\VanDyke Software\VShell\PublicKey\%USER%).
6. Click Apply to save changes.

2.1.8. Starting the VShell Service

1. Start the VShell application.
2. Under Category, click Common.

The Common Server Options page is displayed.



3. Click the Start SSH2 Service button.

The SSH2 Service is now started and can be used by the File Updater to synchronize folders.

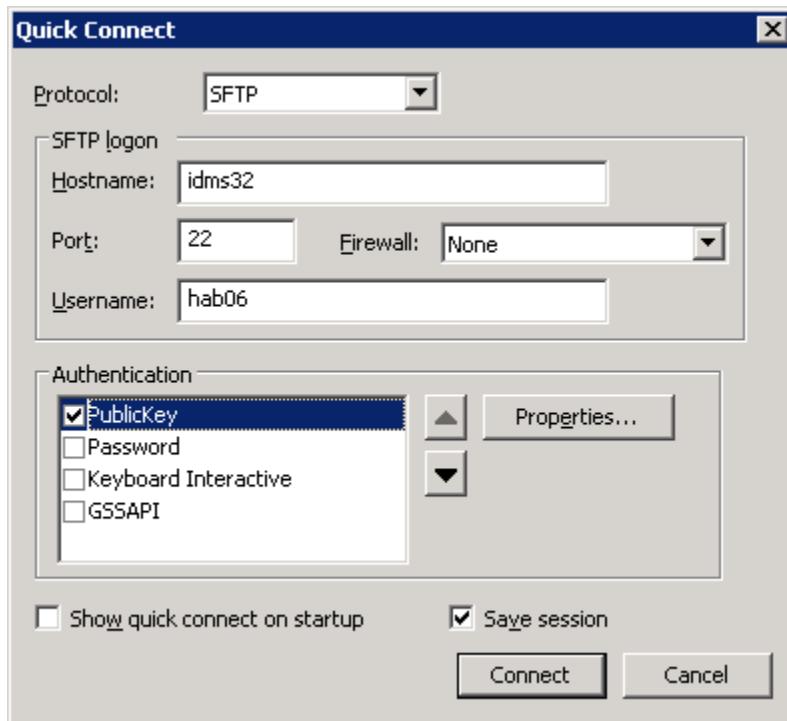
2.1.9. Validating the SSH Connection

1. Start the VanDyke SecureFX application.
2. On the VanDyke SecureFX toolbar, click the Quick Connect button .

The Quick Connect form appears.

3. Enter the parameters as follows:
 - Protocol: SFTP.
 - Hostname: Enter the computer name, or localhost.
 - Port: Use the default value.
 - Firewall: None.
 - Username: Enter the user name.
 -

Authentication: Select the Public Key authentication only.

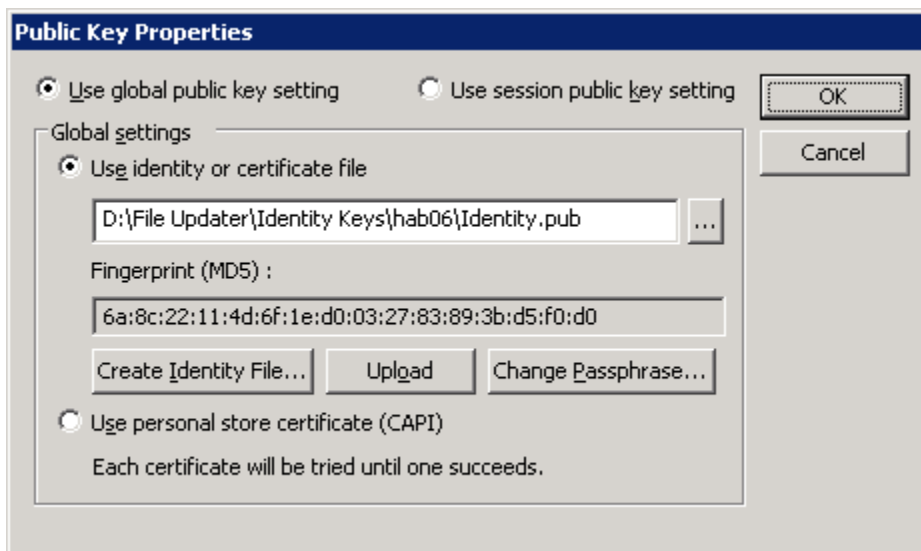
The 'Quick Connect' dialog box is shown. It has a title bar with a close button. The 'Protocol' dropdown is set to 'SFTP'. The 'SFTP logon' section contains fields for 'Hostname' (idms32), 'Port' (22), 'Firewall' (None), and 'Username' (hab06). The 'Authentication' section has a list box with 'PublicKey' selected and checked, and 'Password', 'Keyboard Interactive', and 'GSSAPI' are unchecked. To the right of the list box are 'Properties...' and up/down arrow buttons. At the bottom, there are checkboxes for 'Show quick connect on startup' (unchecked) and 'Save session' (checked), and 'Connect' and 'Cancel' buttons.

4. Under Authentication, select PublicKey and click Properties.

The Public Key Properties form appears.

5. Set the path to the previously generated public key (see [Generating Identity Keys](#)).

Use the default settings for the other options.

The 'Public Key Properties' dialog box is shown. It has a title bar. There are two radio buttons: 'Use global public key setting' (selected) and 'Use session public key setting'. To the right are 'OK' and 'Cancel' buttons. The 'Global settings' section has a radio button 'Use identity or certificate file' (selected). Below it is a text field with the path 'D:\File Updater\Identity Keys\hab06\Identity.pub' and a browse button (...). Below the text field is the 'Fingerprint (MD5) :' label and a text field containing '6a:8c:22:11:4d:6f:1e:d0:03:27:83:89:3b:d5:f0:d0'. Below the fingerprint field are three buttons: 'Create Identity File...', 'Upload', and 'Change Passphrase...'. At the bottom, there is a radio button 'Use personal store certificate (CAPI)' which is unselected. Below it is the text 'Each certificate will be tried until one succeeds.'

6. Click OK to save changes and close the Public Key Properties form.
7. On the Quick Connect form, click Connect to save changes and establish the SFTP connection.

You are connected to the SFTP server without asking for the login and password. Here, you can check the defined SFTP folders (virtual roots).

If you experience any difficulties, refer to the *Secure Shell (SSH) User Guide* for a detailed description of possible solutions.

2.2. FTP Configuration

To use the File Updater via FTP in passive mode, the Microsoft FTP server service must be installed on the File Updater server computer. The File Updater client computer does not require installing any specific component.

This section describes how to deploy the FTP server service for use by the File Updater application.

The deployment procedure has two parts:

- Installing IIS and the FTP service
- Configuring the FTP service

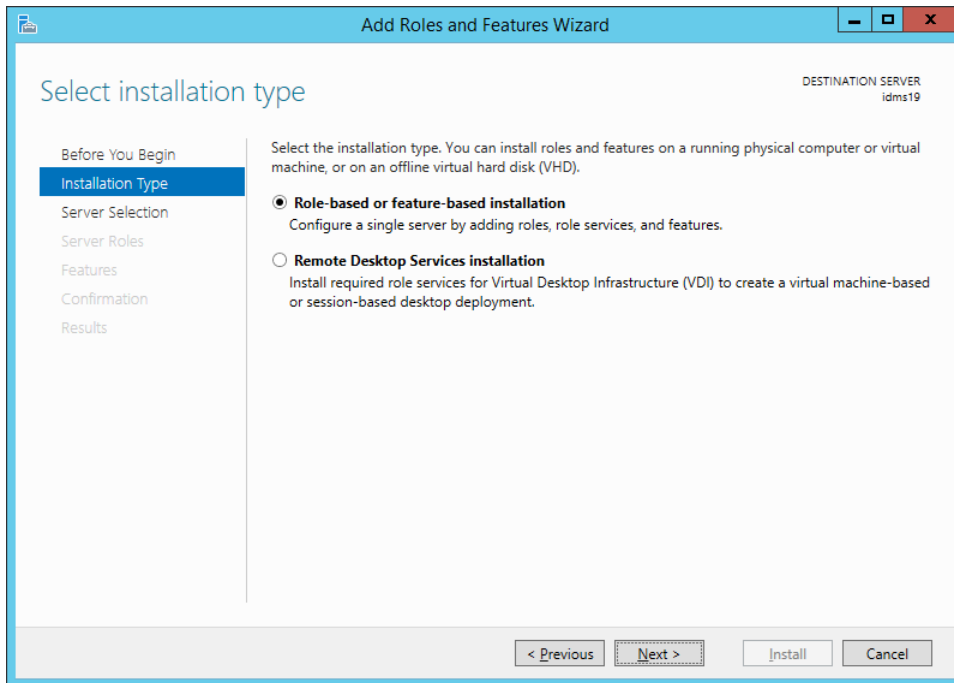
This procedure is required only if FTP is used (not SFTP).

2.2.1. Installing IIS and the FTP Service

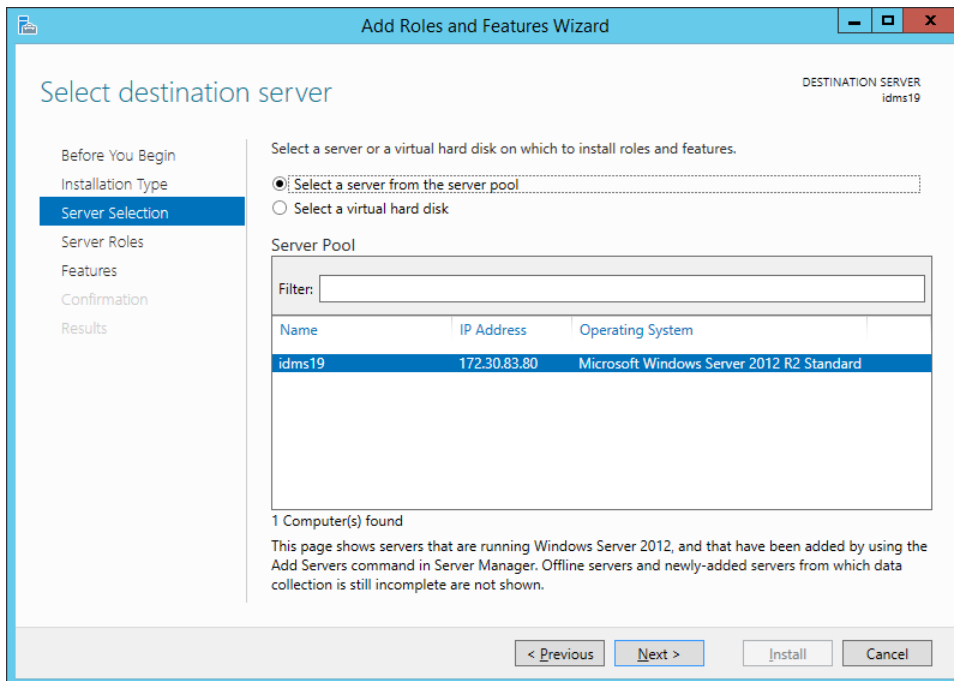
1. On the Windows Start menu, click Server Manager.
2. Click Manage, and then click Add roles and features.

The Add Roles and Features Wizard starts.

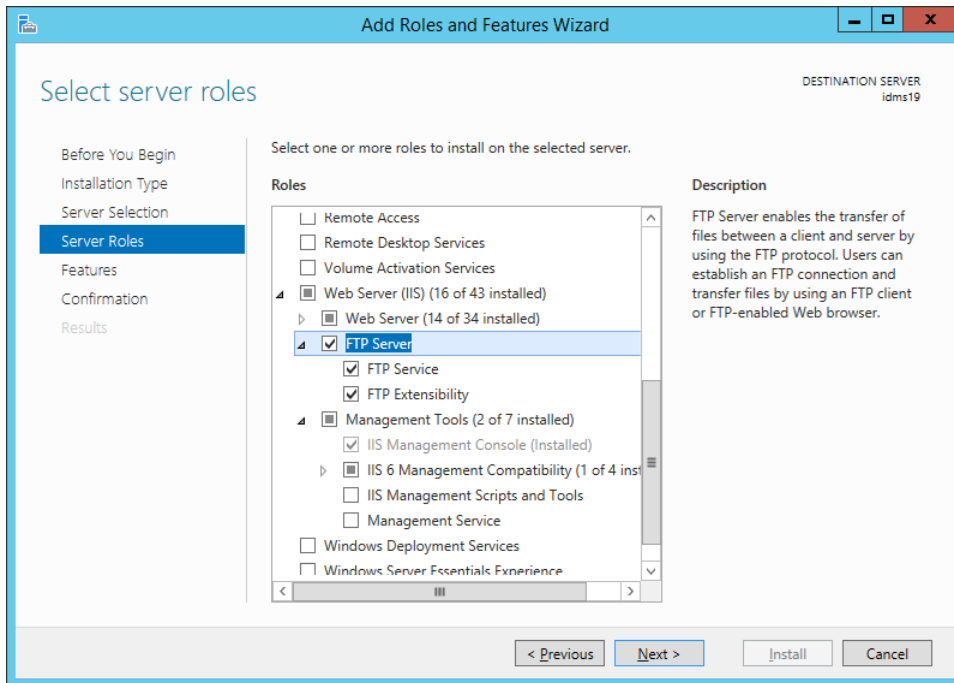
3. Select the "Role-based" installation option, and then click Next.



4. Select the server where you want to install IIS, and then click Next.



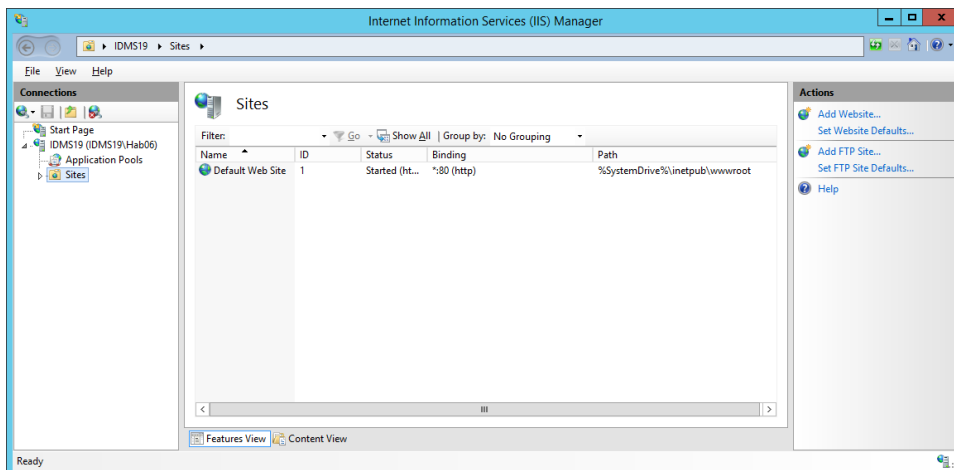
5. Expand Web Server (IIS) and select the FTP Server options as shown below.



6. Click Next.
7. On the Confirm installation selections page, click Install.
8. Click Close.

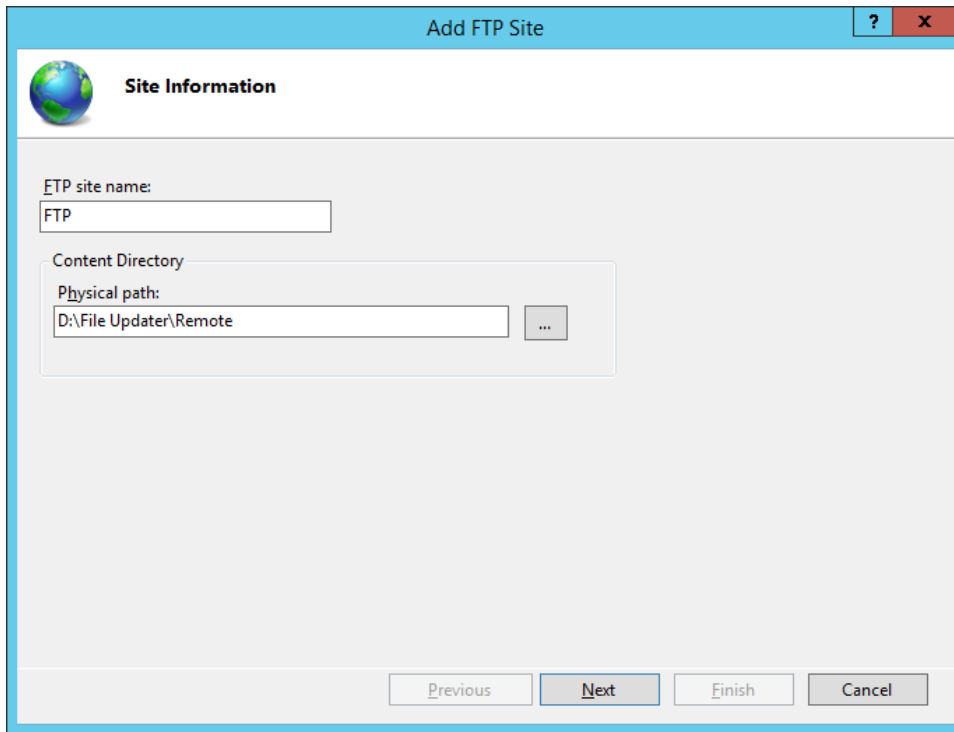
2.2.2. Configuring the FTP Service

1. From the Windows Start menu, click Server Manager.
2. Click tools, and then click Internet Information Services (IIS) Manager.
3. In the Connections pane, expand <your host name>\Sites.



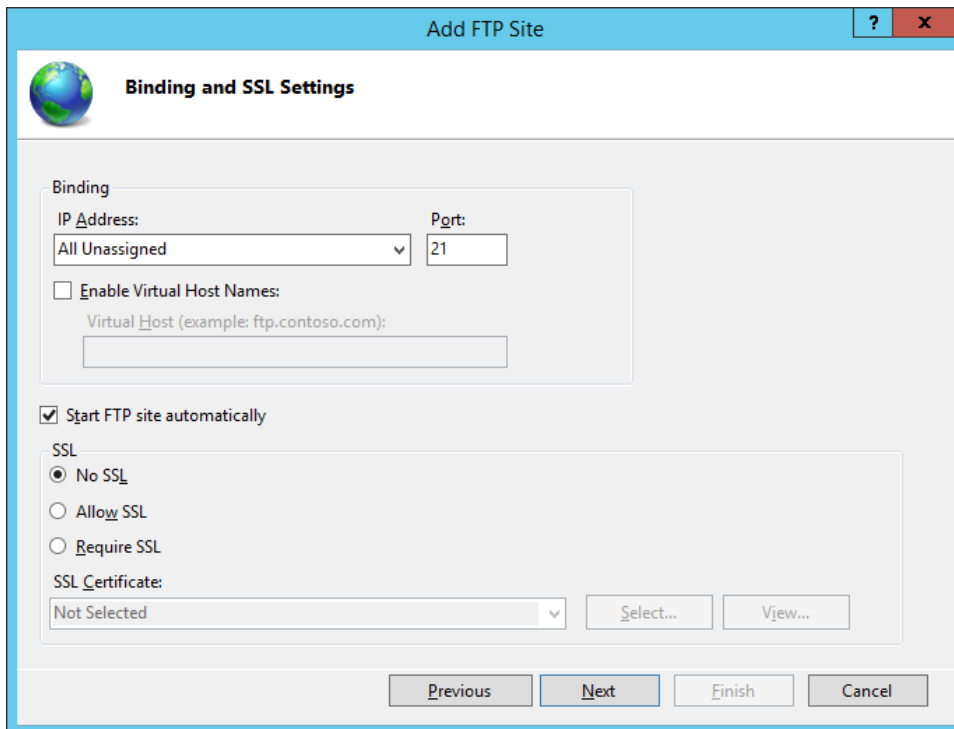
4. In the Actions pane, click Add FTP Site.

5. Specify the FTP site name, enter the path to the FTP folder location, and then click Next.



The screenshot shows the 'Add FTP Site' dialog box with the 'Site Information' tab selected. The 'FTP site name' field contains 'FTP'. The 'Content Directory' section shows the 'Physical path' as 'D:\File Updater\Remote'. At the bottom, there are buttons for 'Previous', 'Next', 'Finish', and 'Cancel'.

6. In the SSL group, select the No SSL option and then click Next.



The screenshot shows the 'Add FTP Site' dialog box with the 'Binding and SSL Settings' tab selected. In the 'Binding' section, 'IP Address' is set to 'All Unassigned' and 'Port' is '21'. The 'Enable Virtual Host Names' checkbox is unchecked. In the 'SSL' section, the 'No SSL' radio button is selected. The 'Start FTP site automatically' checkbox is checked. The 'SSL Certificate' dropdown is set to 'Not Selected'. At the bottom, there are buttons for 'Previous', 'Next', 'Finish', and 'Cancel'.

Note

For instructions on using SSL, refer to the section Enable SSL Support on IIS 8 in the *System Security Guide for Windows Server 2016*.

7. Under Authentication, select Basic.
8. Under Allow access to, select Specified users, and then enter a user name to be used by the File Updater application. Grant the user the Read and Write permissions.

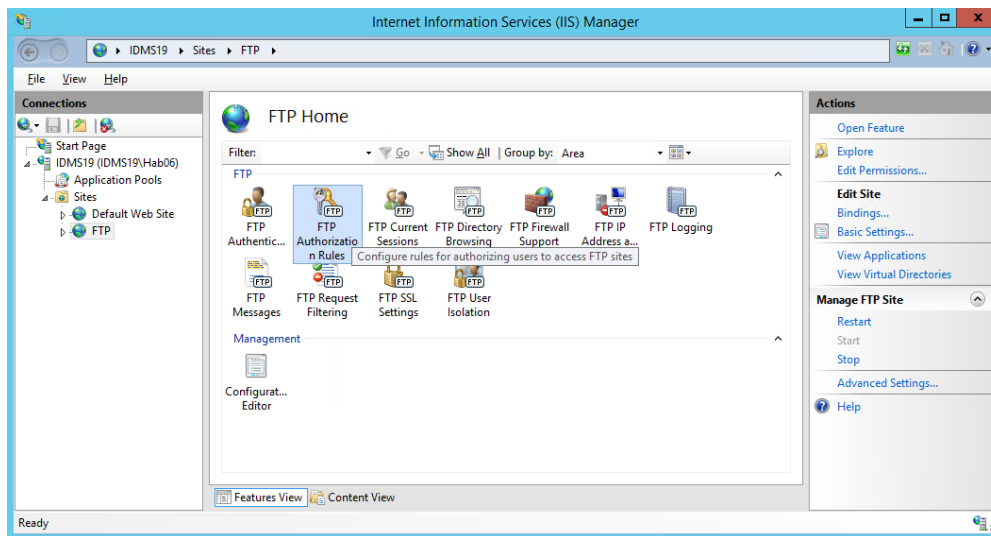
The screenshot shows the 'Add FTP Site' wizard window. The title bar says 'Add FTP Site'. The main content area is titled 'Authentication and Authorization Information'. It contains three sections: 'Authentication' with radio buttons for 'Anonymous' (unchecked) and 'Basic' (checked); 'Authorization' with a dropdown menu set to 'Specified users' and a text box containing 'hab06'; and 'Permissions' with checkboxes for 'Read' (checked) and 'Write' (checked). At the bottom, there are four buttons: 'Previous', 'Next', 'Finish', and 'Cancel'.

9. Click Finish.

The FTP service configuration is complete.

Note

If you need more sophisticated permission settings (for example, one user has read-only access and another user has write access), you can use the default settings now and configure permissions later. You can configure permissions via IIS Manager by choosing the FTP site that you created above and modifying the FTP Authentication and FTP Authorization Rules settings as shown in the image below.



To verify these settings:

1. Connect to the current host using any FTP client with the credentials you specified during setup.
2. Ensure that the specified users have access to the physical path of the FTP server (a folder that is accessible via FTP).

3. Installing the File Updater Server/Client

The ADMS File Updater server/client is normally installed as a Windows service. However, it can also be used as a standalone application.

If an earlier version of the File Updater server/client already exists on the computer, it must be uninstalled.

- To uninstall the ADMS File Updater server, run the file `ADMS_fileupdaterserver_X64.msi`.
- To uninstall the ADMS File Updater client, run the file `ADMS_fileupdaterclient_X64.msi`.

To prevent installation issues with newer versions of the File Updater software, GE recommends removing the log files folder.

To install the File Updater server/client:

- To install the ADMS File Updater server, run the file `ADMS_fileupdaterserver_X64.msi`.
- To install the ADMS File Updater client, run the file `ADMS_fileupdaterclient_X64.msi`.

Note

All installation and configuration parameters defined must contain only ASCII characters.

To change the startup type:

1. Run `services.msc` from the Windows Start menu or command line.

The Service Management Console opens.

2. Locate the ADMS File Updater server and ADMS File Updater client services.
3. Right-click the service provider and select Properties.

The Properties dialog box appears.

4. On the General tab, change the Startup Type to Automatic.
5. Click Apply and then click OK.

4. Configuring the File Updater Server/Client

4.1. File Updater Configuration Editor Overview

To start the File Updater Configuration Editor, click Start > Programs > Eterra > Distribution > File Updater Configuration Editor.

The following image shows the File Updater Configuration Editor's main window with a loaded File Updater server configuration.

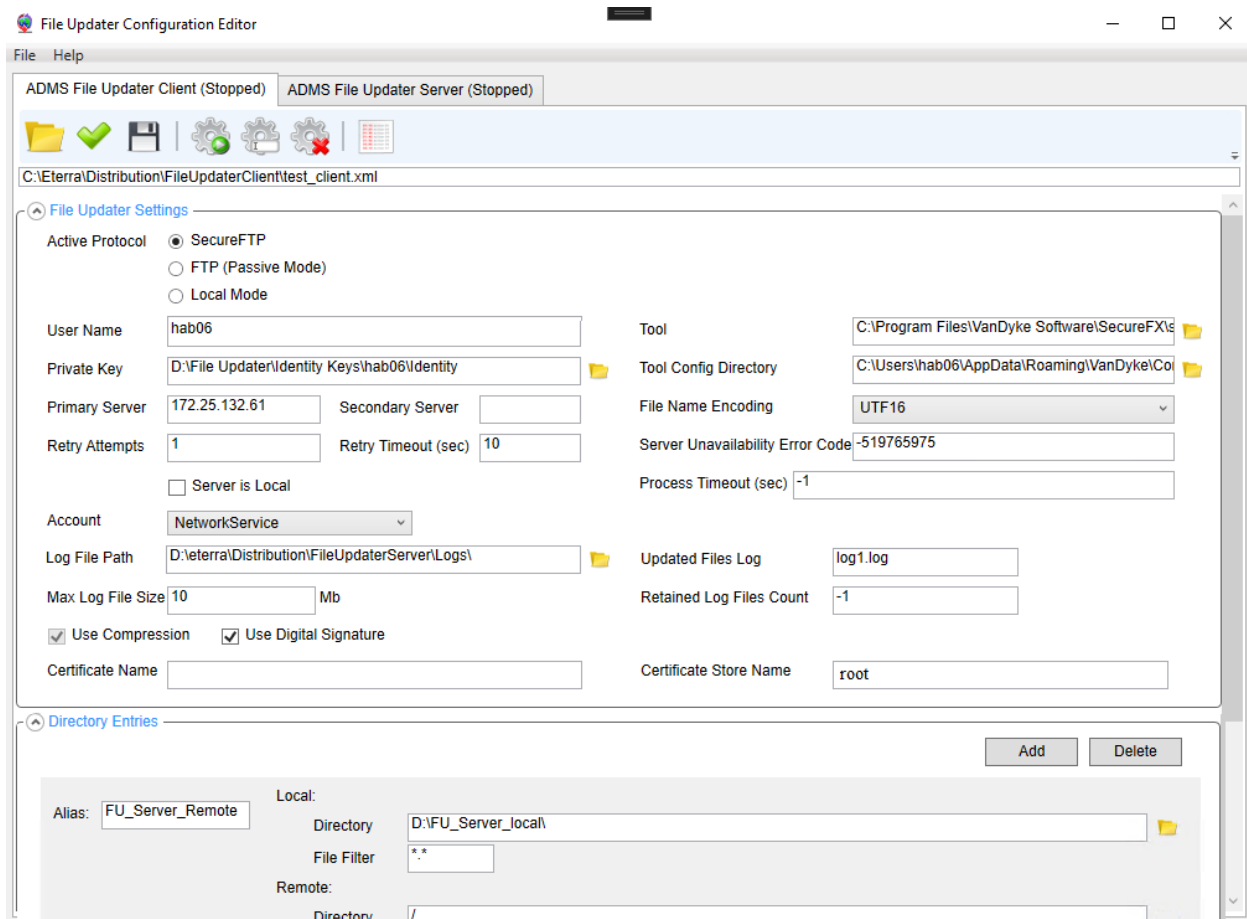


Figure 4. File Updater Configuration Editor

Upon startup, the File Updater Configuration Editor automatically detects installations of the File Updater server and the File Updater client and loads the configuration file(s). The Configuration Editor tool opens a tab for every instance of a detected service.

Note

File Updater version 2.5.4 can apply configuration files of the same version only. File Updater version 3.0.1 and later are completely compatible with configuration files of versions 3.0.1 and later. To convert a configuration file from version 2.5.4 to a later version, open it with a File

Updater of a later version, and save it. The file obtains the supported xml file structure.

Every tab has its own toolbar that applies only to the corresponding File Updater service instance. The following options are available on the toolbar.



- **Open Configuration File:** Click to open a new configuration file.
- **Validate Current Settings:** Click to validate the current configuration.

Validating before saving the configuration is recommended. The validation details are described in the Validation Log section. Any incorrect fields are outlined in red.

- **Save:** Click to save the current configuration. This step also updates the File Updater service parameters with the configuration file name.
- **Start/Stop Service:** Click to start or stop the corresponding service. You should validate and save the configuration before starting the service.
- **Rename Service:** Click to rename the service.
- **Delete Service:** Click to remove the service from the File Updater Configuration Editor and the Windows services list.
- **Open Log:** Click to open the File Updater log file for the current date. If a log for the current date does not exist, the Log directory is opened.

The text field below the toolbar shows the path to the currently loaded configuration file.

D:\Eterra\Distribution\FileUpdaterServer\fusrv_conf.xml

The Configuration Editor display contains three sections:

- The File Updater Settings section defines the protocol parameters and main service paths. For more information, see [FTP Configuration](#) and [SecureFTP Configuration](#).
- The Directory Entries section defines how directories are synchronized. For details, see [Directory Synchronization](#).
- The Validation Log section shows configuration errors and warnings. For more information, see [Validation](#).

Common configuration settings are:

- **Active Protocol:** Specifies which file transfer protocol to use.
 - **FTP:** File Updater uses an FTP server to synchronize files.
 - **SecureFTP:** File Updater uses a SecureFX server to synchronize files.
 - **Local Mode:** File Updater uses its own capabilities to synchronize files. This option can only be used when the local and remote folders are located on the same host. For example, you may configure the File Updater Server to operate in the Local mode and File Updater Client

operate in the FTP or SecureFTP mode. The local mode does not require installing third-party software, such as SecureFX server.

After you select a protocol, a corresponding configuration section becomes visible. For more details, see [FTP Configuration](#), [SecureFTP Configuration](#), and [Local Mode Configuration](#).

- **Account:** Select this option to change the logon account for the File Updater application(s). The "NetworkService" and "LocalSystem" services are supported. By default, the service always logs on as "NetworkService". For more information, see [Terminology](#).

Note

To change the logon account, see [Services](#).

- **Log File Path:** Sets the full path to a location where log files will be saved. The File Updater server should have permissions to write to the specified folder.

Note

In SFTP mode, the log file path must not contain spaces due to restrictions of the SecureFX command-line tool, sfxcl.

- **Updated Files Log:** Sets the name for the Updated Logs file. For more information, see [Terminology](#).
- **Max Log File Size:** When the current log file reaches the specified size, the file is renamed and a new log file is created.
- **Retained Log Files Count:** When the number of log files in the log folder exceeds the specified number, the oldest log file is retired. If set to "-1", all log files are retained in the log folder.
- **Use Compression:** Select this option to enable file compression. If this option is used, the synchronized files are compressed to gzip format (.gz).
- **Use Digital Signature:** Select this option to apply a digital signature to each file that is synchronized. If this option is used, each synchronized file is digitally signed using the certificate name as specified in "Certificate Name" and installed in the certificate store identified by "Certificate Store Name".
- **Certificate Store Name:** Contains the name of the certificate store in which the certificate was installed.
- **Certificate Name:** Contains the name of the certificate to use for digitally signing files.

Note

To avoid errors, the Use Compression option must be set similarly on the File Updater server and client applications.

Note

If compression is disabled, and UTF16 is enabled, files are uploaded to the remote server in

a batch, "as is", with their original file names.

Note

It is highly recommended to install digital certificates in the local machine in the "root" certificate store, also known as Trusted Root Certification Authority.

Note

If used, the digital signature must be configured appropriately in all server and client instances. If enabled in the server, it must be enabled in the clients.

Note

Digital Signatures functionality is only available in Windows.

4.2. FTP Configuration

File Updater Settings

Active Protocol: ☐ SecureFTP ☒ **FTP (Passive Mode)** ☐ Local Mode

User Name:

Password:

Repeat Password:

Server: Port:

☐ Server is Local

Log File Path:

Updated Files Log:

Max log file size (MB):

Retained log files count:

☐ Use Compression

Figure 5. FTP Configuration

The following configuration options are specific to the FTP mode:

- **User Name:** Contains login information for a user registered on the FTP server (see [FTP Configuration](#)).
- **Password:** Contains the user's password (Windows user password). The password is used only for the FTP connection. It is encoded automatically (see [Password Encryption](#)).

If you selected anonymous authentication for the FTP server (see [FTP Configuration](#)), the password is not required.

- **Server:** Contains the server name or IP address of the FTP server.
- **Port:** Contains the port number of the FTP server; the default value is 21.
- **Server Is Local:** If selected, the value in the Server field is ignored and the IP address 127.0.0.1

(localhost) is used.

4.2.1. Password Encryption

The File Updater automatically stores passwords as an encoded string. This is highly recommended for security.

To encode the FTP password manually, run the following command:

```
IDMS_FileUpdaterServer.exe /IDMS_FileUpdaterClient.exe /SETPASSWORD conf.xml ftp_password
```

where:

conf.xml is the configuration file where the encoded password is set.

ftp_password is the password to be encoded.

After running this command, the File Updater updates the Password field with the encoded value. Encoded values end with "ENCODED".

For backward compatibility, the File Updater can read passwords stored as clear text.

4.3. SecureFTP Configuration

File Updater Settings

Active Protocol: ☒ SecureFTP, ☐ FTP (Passive Mode), ☐ Local Mode

User Name: hab06

Private Key: D:\File Updater\Identity Keys\hab06\Identity

Primary Server: 1.2.3.4.5, Secondary Server:

Retry Attempts: 1, Retry Timeout (sec): 10

☐ Server is Local

Log File Path: d:\Eterra\Distribution\FileUpdaterServer\Logs\

Max log file size (MB): 10

☐ Use Compression

Tool: C:\Program Files\VanDyke Software\SecureFXs

Tool Config Directory: C:\Users\hab06\AppData\Roaming\VanDyke\cor

File Name Encoding: UTF16

Server Unavailability Error Code: -519765975

Process Timeout (sec): -1

Updated Files Log: UpdatedFilesLog.xml

Retained log files count: -1

Figure 6. SecureFTP Configuration

The following configuration options are specific to the SecureFTP mode:

- **User Name:** The login name of a user registered on the SFTP server (see [SFTP Installation and Configuration](#)).
- **Private Key:** For each File Updater instance, set to the full path of the generated private key (see [Generating Identity Keys](#)).
- **Primary Server, Secondary Server:** The host name or the IP address of a VShell server to use.
- **Retry Attempts:** The number of connection retries in case an error occurs while attempting to

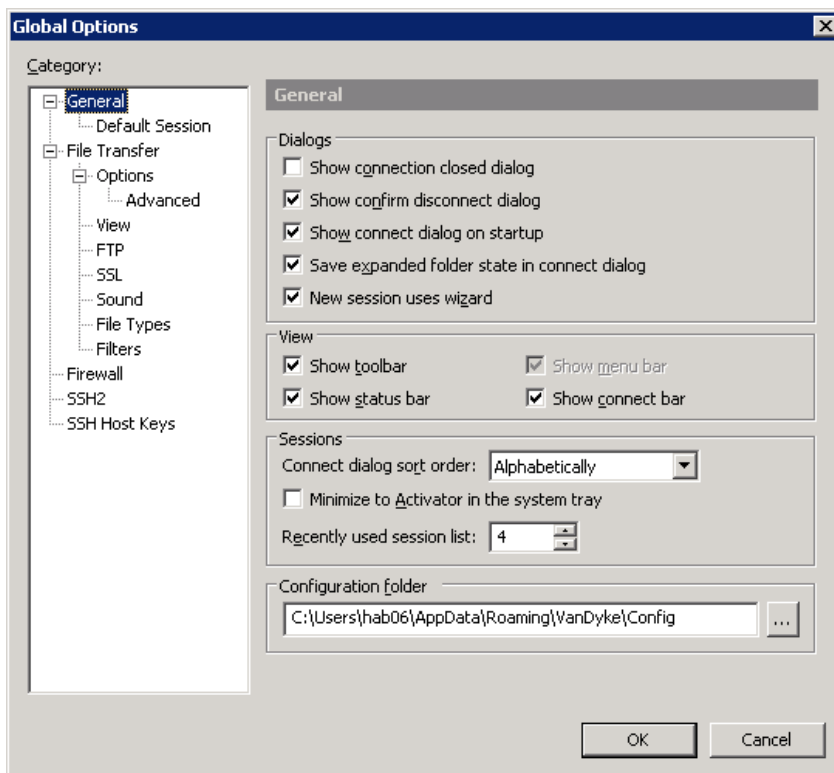
connect.

- **Retry Timeout:** The interval (in seconds) between connection retries in case an error occurs while attempting to connect.
- **Server Is Local:** If selected, both primary and secondary servers are ignored and the IP address 127.0.0.1 (localhost) is used.
- **Tool:** The full path to the VanDyke SSH command-line executable file (C:\Program Files\VanDyke Software\SecureFX\sfxcl.exe). This path is used as a configuration directory of VanDyke's SFTP client if the Tool Config Directory parameter is not specified. Therefore, it is important that the executable file is located in its installation directory (where it was created by the installer).

This parameter is used only in Windows.

- **Tool Config Directory:** The full path to the configuration directory of VanDyke's SSH command-line tool (C:\Users\<User>\AppData\Roaming\VanDyke\Config). The setting should match the value set in the Configuration Folder text field on the General tab in SecureFX's Global Options window.

If no path is specified, File Updater assumes this value is the Config subfolder of the directory specified in the Tool parameter.



This parameter is used only in Windows. The setting cannot contain international characters.

- **Server Unavailability Error Code:** This value (integer) is used by the "server switch" functionality. This parameter is used only in Windows with SFTP.

The SecureFX utility has a specific error code for cases when the SFTP server is unavailable

(VShell or the server host itself). Usually, this error code value matches the Server Unavailability Error Code parameter in the generated configuration file. When the error code is returned from SecureFX, it tells File Updater about the SFTP server unavailability.

It is recommended that you use the default value of the Server Unavailability Error Code parameter provided in the generated (new) configuration file. If the default value does not work as expected, locate the expected Error Code value using the steps below:

- Configure the File Updater to use the unavailable SFTP server.
- Initiate the file transfer.
- In the log file, locate a failed operation message entry with the expected error code value. For example:

```
"Connection test to the server Host2 failed, error code: -519765975 while server unavailability error code is: -100"
```

- **File Name Encoding:** Determines the encoding in which file names are sent to the SFTP server. UTF8 and UTF16 are supported.
- **Process Timeout:** Sets the time period (in seconds) to complete each file processing. If a file is not uploaded within the defined period, the upload process is terminated by the application. This option is used to clean up stale processes. The default value "-1" defines that the process is never terminated due to timeout.

4.3.1. Server Switch Feature

If the SFTP server host is unavailable, the File Updater client application switches to another SFTP server (if a secondary server is defined in the configuration file).

In the configuration file, two server IP addresses are specified as "Primary" and "Secondary". By default, the IP address specified as "Primary" is used. If an attempt to connect to the SFTP server fails, the IP address specified as "Secondary" is used for the next synchronization iteration.

Every time the File Updater client identifies an unavailability error, it switches to an alternative IP address for SFTP (primary or secondary). This continues until the File Updater client successfully establishes a connection to the SFTP server.

Switching to a different server only occurs in the case of host unavailability, or if the File Updater client is unable to connect to the SFTP server service. No other errors (authorization failed, file not found on server, etc.) cause a server switch.

Using the Server Switch feature for the File Updater server is deprecated.

Note the following:

- The Server Switch functionality is not supported for the FTP connection.
- When a server switch happens on the File Updater server side, it clears the file information registry (UpdatedFilesLog.xml) and performs a full update for a new server. Information about

file deletion is not preserved. If clients do not synchronize file deletion information before the server switch occurs, the file to be deleted stays on the clients' side.

4.3.2. Redundancy Configuration

The following image demonstrates the File Updater redundancy configuration model where the File Updater clients use the Server Switch feature and two instances of the File Updater server. Note that servers do not use the Server Switch functionality.

In this model, the redundancy configuration includes:

- A backup SFTP server host, SFTP Server 2.
- Two modeling hosts equipped with an instance of the File Updater server connected to a separate SFTP server.
- File Updater clients.

In this model, the clients use SFTP Server 1. If SFTP Server 1 becomes unavailable, the clients switch to SFTP Server 2 and download only the changed files.

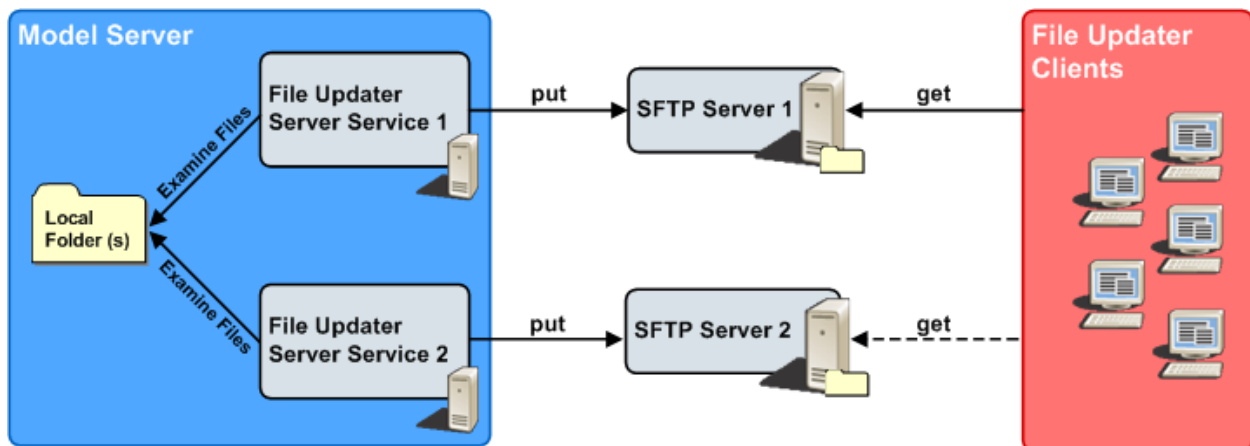


Figure 7. Redundancy Configuration

4.4. Local Mode Configuration

File Updater Settings

Active Protocol: ☐ SecureFTP ☐ FTP (Passive Mode) ☒ Local Mode

Remote root:

Log File Path:

Max log file size (MB):

Updated Files Log:

Retained log files count:

Figure 8. Local Mode Configuration

The following configuration option is specific to the Local mode:

- **Remote Root:** Absolute path to a shared folder used to store remote folders specified by the "Remote Directory" parameters. This folder must be located on the local host.

4.5. Directory Synchronization

The File Updater server and the File Updater client allow you to define any number of folders to be synchronized. It is possible to define any relationship between local and remote folders (for example, one-to-one, many-to-many, etc.).

One directory entry defines a one-to-one relationship between a local folder and a remote folder. To define multiple remote folders to synchronize with one local folder, you are required to specify as many directory entries as needed to define each local and remote folder mapping. The same approach is needed for defining multiple local folders to be synchronized with one remote folder.

4.5.1. Using Configuration Editor to Define Synchronized Directories

Figure 9. Defining Synchronized Directories

- **Alias:** Brief directory alias name that is used in the log file to distinguish records that are related to different directories. The alias name is also specified when configuring VShell Virtual Roots (see [Configuring the VShell Virtual Roots \(Defining SFTP Folders\)](#)).
- **Local Directory:** Absolute patch to a local folder that is synchronized with a remote folder, specified in the Remote Directory node.
- **File Filter:** Defines file name patterns to specify which files are to be synchronized. Several wildcards can be defined. Wildcards are separated with a semi-colon (";") character (for example, *.bmp*;*.xml*).
- **Remote Directory:** Relative path to a folder inside a remote folder on the FTP/SFTP server or local host (for the Local mode). Files are downloaded from this location (for File Updater client) or uploaded to this location (File Updater server). Paths should be valid according to the

FTP/SFTP server host.

Note

Do not include the absolute path to the FTP/SFTP folder on the FTP/SFTP server computer in the Remote Directory field. For instance, for the Models folder, located on the SFTP server (D:\File Updater\Remote) at D:\File Updater\Remote\Models, enter "/Models" or just "Models".

Note

The value of this parameter is case-sensitive. Ensure that it is the same in both the File Updater server and client configuration files.

- **Continue on Error:** Specifies whether the client/server should continue synchronizing this folder if any errors occur. When selected, the client/server continues synchronizing even if errors occur. If not selected, the client/server aborts synchronizing and undoes all actions when an error occurs.
- **Time Delay (Primary Comparison):** Defines the time period (in seconds) before checking the current directory for changes based on the file size and last modified time after the previous synchronization task has been completed.
- **Time Delay (Extended Comparison):** Defines the time period (in seconds) before checking the current directory for changes based on the file CRC, size, and last modified time after the previous synchronization task has been completed.

Note

The CRC calculation is an extensive task, which might cause performance issues.

- **No Synchronization Message Needed Interval:** Defines the time period (in seconds) for the "No data to synchronize" message to appear in the log. For the "0" value, the message appears only after the service start or last successful synchronization. The message is repeated as per the defined value, if it is greater than the Time Delay value. For a value less than the Time Delay value, the message is repeated as per the Time Delay value.

4.5.2. Configuration Examples

The configuration examples below demonstrate directory entry definitions for each type of relationship between local folder(s) and remote folder(s).

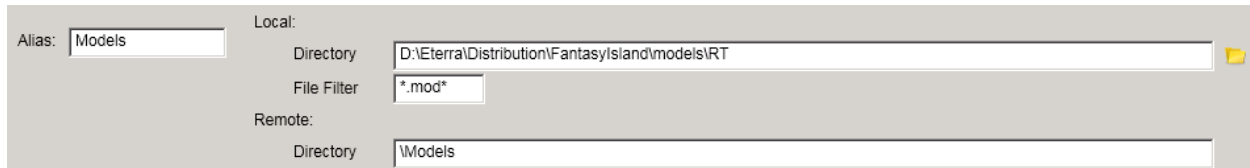
Note

Remote folder names are case-sensitive, which may cause problems. Remote folder names that differ only in case (for example, /ftpdire and /ftpdIR) are considered different folders. Be sure to name the remote folders in the File Updater client and File Updater server configuration files

identically.

4.5.2.1. One-To-One: One Local Folder and One Remote Folder

In this example, the remote folder \Models is synchronized with the local folder D:\Eterra\Distribution\FantasyIsland\models\RT.

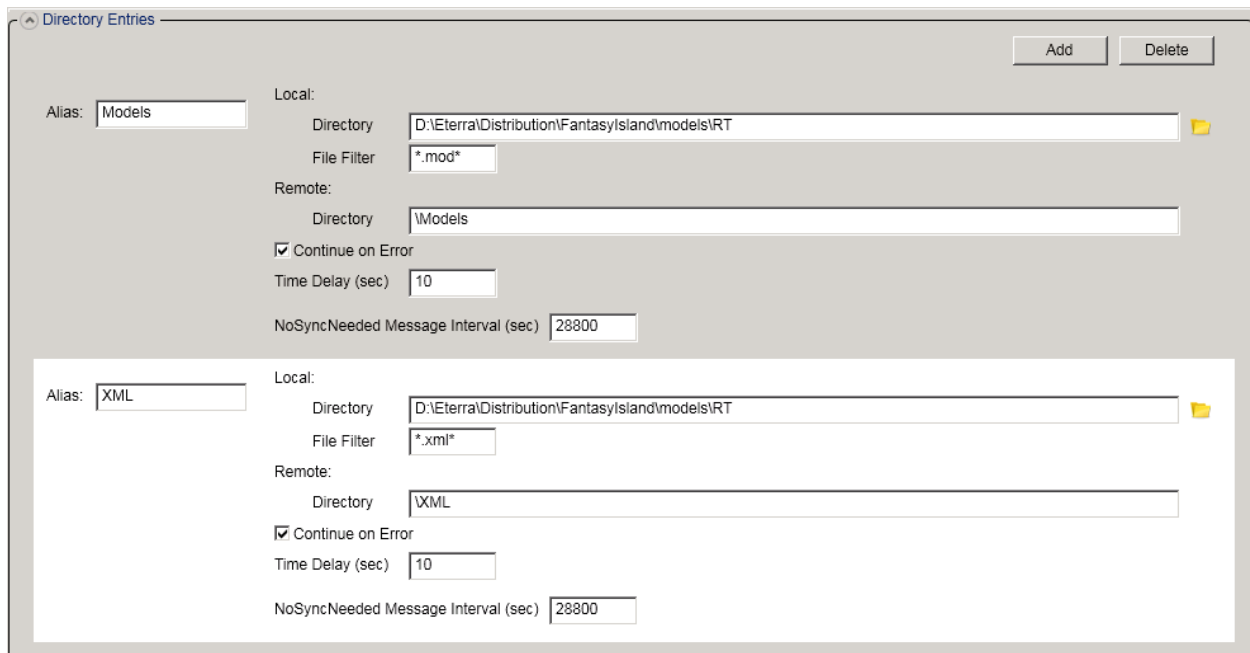


The screenshot shows the 'Directory Entries' configuration window. It has two main sections. The top section is for an entry with Alias 'Models'. The 'Local' section has 'Directory' set to 'D:\Eterra\Distribution\FantasyIsland\models\RT' and 'File Filter' set to '*.mod*'. The 'Remote' section has 'Directory' set to '\Models'. There are 'Add' and 'Delete' buttons at the top right.

Figure 10. Example of a One-To-One Relationship

4.5.2.2. One-To-Many: One Local Folder and Many Remote Folders

In this example, remote folders \Models and \XML are synchronized with the same local folder, D:\Eterra\Distribution\FantasyIsland\models\RT.



The screenshot shows the 'Directory Entries' configuration window with two entries. The top entry has Alias 'Models', Local Directory 'D:\Eterra\Distribution\FantasyIsland\models\RT', File Filter '*.mod*', and Remote Directory '\Models'. The bottom entry has Alias 'XML', Local Directory 'D:\Eterra\Distribution\FantasyIsland\models\RT', File Filter '*.xml*', and Remote Directory '\XML'. Both entries have 'Continue on Error' checked, 'Time Delay (sec)' set to 10, and 'NoSyncNeeded Message Interval (sec)' set to 28800. 'Add' and 'Delete' buttons are at the top right.

Figure 11. Example of a One-To-Many Relationship

4.5.2.3. Many-To-One: Many Local Folders and One Remote Folder

In this example, the remote folder \Models is synchronized with the local folders D:\Eterra\Distribution\FantasyIsland\models\RT and D:\Eterra\Distribution\FantasyIsland\models\ST.

The screenshot shows a window titled "Directory Entries" with two tabs. The first tab, "Models RT", has the following settings: Alias: Models RT, Local Directory: D:\EternalDistribution\FantasyIsland\models\RT, File Filter: *.mod*, Remote Directory: \Models, Continue on Error: checked, Time Delay (sec): 10, and NoSyncNeeded Message Interval (sec): 28800. The second tab, "Models ST", has identical settings: Alias: Models ST, Local Directory: D:\EternalDistribution\FantasyIsland\models\ST, File Filter: *.mod*, Remote Directory: \Models, Continue on Error: checked, Time Delay (sec): 10, and NoSyncNeeded Message Interval (sec): 28800. Buttons for "Add" and "Delete" are in the top right corner.

Figure 12. Example of a Many-To-One Relationship

4.5.3. Defining Dependent Folders

The File Updater server and the File Updater client allow dependent folders (in terms of synchronization) to be defined. This means if any dependent folders synchronize with errors, all of the dependent folders are resynchronized from the beginning. Dependent folders are successfully synchronized only when all of the dependent folders are successfully synchronized. This feature allows atomic updates to several folders.

For example, you may use dependent folders when you need to add a new display and bitmaps to a cross-referenced displays folder and images folder. To correctly process the new display, the display and image files must be available.

Using dependent folders allows you to prevent situations where the displays folder is updated but the images folder update fails because of a network error, which results in an inconsistent display without images.

To configure dependent directory entries, you need to enter equal values in the Time Delay field as shown in the following image. In this example, the first two directory entries ("Models RT" and "Models ST") are dependent, while the last directory entry ("Bitmaps") is independent.

! Important

Unless it is necessary, do not define the same value of the Time Delay attribute unless the directory entries are dependent.

The screenshot shows the 'Directory Entries' window with three entries configured:

- Models RT:**
 - Alias: Models RT
 - Local Directory: D:\Eterra\Distribution\FantasyIsland\models\RT
 - Local File Filter: *.mod*
 - Remote Directory: \Models
 - ☒ Continue on Error
 - Time Delay (sec): 10
 - NoSyncNeeded Message Interval (sec): 28800
- Models ST:**
 - Alias: Models ST
 - Local Directory: D:\Eterra\Distribution\FantasyIsland\models\ST
 - Local File Filter: *.mod*
 - Remote Directory: \Models
 - ☒ Continue on Error
 - Time Delay (sec): 10
 - NoSyncNeeded Message Interval (sec): 28800
- Bitmaps:**
 - Alias: Bitmaps
 - Local Directory: D:\Eterra\Distribution\FantasyIsland\bitmaps
 - Local File Filter: *.bmp*,*.png
 - Remote Directory: \Bitmaps
 - ☐ Continue on Error
 - Time Delay (sec): 30
 - NoSyncNeeded Message Interval (sec): 28800

Figure 13. Configuring Dependent Folders

4.6. Manually Editing the Configuration XML File

Warning

GE does not recommend editing the XML file manually! If you still wish to do so, always run a validation before applying the modified configuration file.

Warning

This XML structure should not be changed. Only node values should be edited.

The File Updater server configuration is represented by an XML file. By default, it is named `fusrv_conf.xml` and located in the same folder as the File Updater server executable file, in `D:\Eterra\Distribution\FileUpdaterServer`.

To configure the File Updater server, the support engineer may edit this file with an XML-based editor.

The File Updater server configuration file has the following structure:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
<!DOCTYPE boost_serialization>
<boost_serialization signature="serialization::archive" version="11">
<pSettings class_id="0" tracking_level="0" version="4">
  <sAccount> NetworkService</sAccount>
  <sLogFilePath>d:\Eterra\Distribution\FileUpdaterServer\Logs\</sLogFilePath>
  <sLogFileType>html</sLogFileType>
  <sUseCompression>1</sUseCompression>
  <sFileTransferProtocol>FTP</sFileTransferProtocol>
  <mFTPSettings class_id="1" tracking_level="0" version="0">
    <sUserName>anonymous</sUserName>
    <sPassword></sPassword>
    <bServerIsLocal>0</bServerIsLocal>
    <sServer>localhost</sServer>
    <sPort>21</sPort>
  </mFTPSettings>
  <mSecureFTPSettings class_id="2" tracking_level="0" version="2">
    <sUserName></sUserName>
    <sPrivateKey></sPrivateKey>
    <bServerIsLocal>0</bServerIsLocal>
    <sPrimaryServer></sPrimaryServer>
    <sSecondaryServer></sSecondaryServer>
    <iRetry>1</iRetry>
    <iRetryTimeout>10</iRetryTimeout>
    <sTool></sTool>
    <sToolConfigDir></sToolConfigDir>
    <sFileNameEncoding>UTF16</sFileNameEncoding>
    <iServerUnavailabilityErrCode>-519765975</iServerUnavailabilityErrCode>
    <iProcessTimeout>-1</iProcessTimeout>
  </mSecureFTPSettings>
  <sUseDigitalSignature>1</sUseDigitalSignature>
  <sCertStoreName>root</sCertStoreName>
  <sSignerName>www.example.com</sSignerName>
  <sLogFileSize>10</sLogFileSize>
  <sLogFileCount>-1</sLogFileCount>
  <mLocalModeSettings class_id="3" tracking_level="0" version="0">
    <sRemoteRoot></sRemoteRoot>
  </mLocalModeSettings>
  <sUpdatedFilesLog>UpdatedFilesLog.xml</sUpdatedFilesLog>
  <mDirectories class_id="4" tracking_level="0" version="0">
    <count>3</count>
    <item_version>3</item_version>
    <item class_id="5" tracking_level="0" version="3">
      <Alias>models</Alias>
      <RemoteDir></RemoteDir>
      <LocalDir>D:\upload\model</LocalDir>
      <Wildcards>*.mod</Wildcards>
      <TimerDelay>10</TimerDelay>
      <sContinueOnError>0</sContinueOnError>
      <NoSyncNeededMessageInterval>28800</NoSyncNeededMessageInterval>
      <TimerDelayCRC>300</TimerDelayCRC>
    </item>
  </mDirectories>
</pSettings>
```

```

</item>
<item>
  <Alias>customer</Alias>
  <RemoteDir></RemoteDir>
  <LocalDir>D:\upload\customer</LocalDir>
  <Wildcards>*.bpcl</Wildcards>
  <TimerDelay>10</TimerDelay>
  <sContinueOnError>0</sContinueOnError>
  <NoSyncNeededMessageInterval>28800</NoSyncNeededMessageInterval>
  <TimerDelayCRC>300</TimerDelayCRC>
</item>
<item>
  <Alias>crew</Alias>
  <RemoteDir></RemoteDir>
  <LocalDir>D:\upload\crew</LocalDir>
  <Wildcards>*.Crew</Wildcards>
  <TimerDelay>10</TimerDelay>
  <sContinueOnError>0</sContinueOnError>
  <NoSyncNeededMessageInterval>28800</NoSyncNeededMessageInterval>
  <TimerDelayCRC>300</TimerDelayCRC>
</item>
</mDirectories>
</pSettings>
</boost_serialization>

```

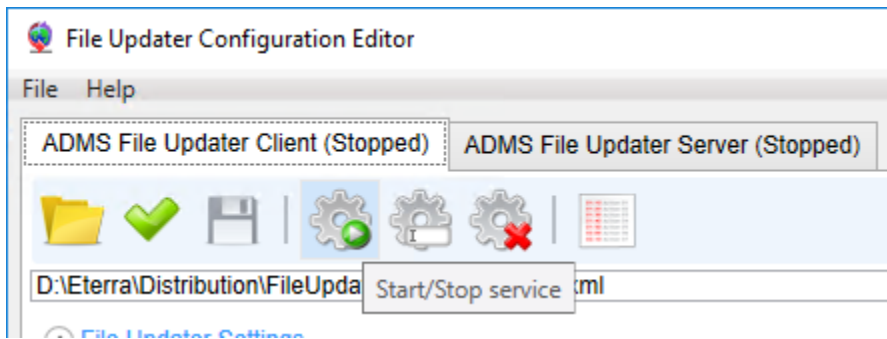
5. Using the File Updater

5.1. Starting the File Updater Server/Client

To create a new File Updater Client or File Updater Server service, select the corresponding option via the File menu.

To start the File Updater server/client service:

- In the Configuration Editor, click the Start Service toolbar button.



Or:

- Use the following shortcut: Start > Programs > Eterra > Distribution > File Updater > Start FileUpdater server/client.

Or:

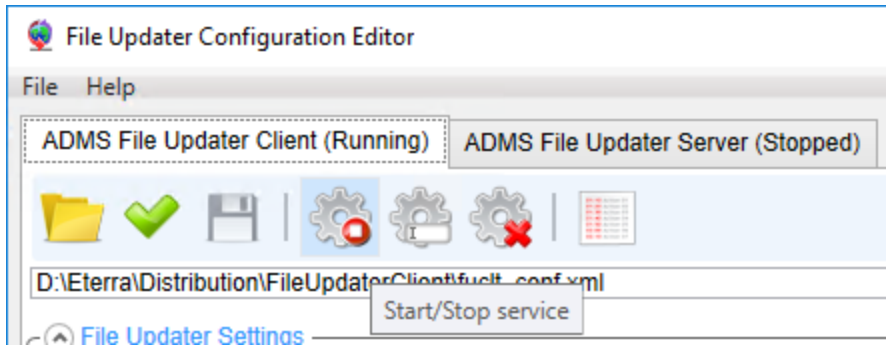
1. Open the Windows Control Panel > Administrative Tools > Services.
2. Select the service named "ADMS File Updater Server"/"ADMS File Updater Client".
3. Click the Start Service button.

When the File Updater server starts it synchronizes all folders that are defined in the configuration file, regardless of the defined Time Delay value. However, the Time Delay value is used in subsequent synchronizations.

5.2. Stopping the File Updater Server/Client

To stop the File Updater server/client service:

- In Configuration Editor, click the Stop Service toolbar button.



Or:

- Use the following shortcut: Start > Programs > Eterra > Distribution > File Updater > Stop FileUpdater server/client.

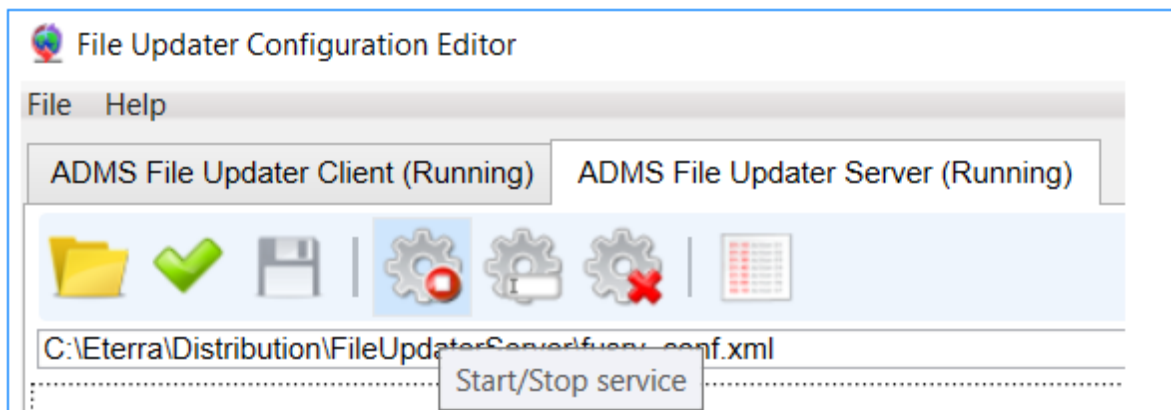
Or:

1. Open the Windows Control Panel > Administrative Tools > Services.
2. Select the service named "ADMS File Updater Server"/ "ADMS File Updater Client".
3. Click the Stop Service button.

5.3. Updating Logon Account for the File Updater Server/Client Services

To update Logon Account for the File Updater server/client service:

1. In the Configuration Editor, click the Stop Service toolbar button.



2. Select the LocalSystem option from the Account dropdown menu.

Note

The default selection is NetworkService.

File Updater Settings

Active Protocol: ☐ SecureFTP
☐ FTP (Passive Mode)
☒ Local Mode

Remote root: C:\Eterra\Distribution\FileUpdaterServer\Remote

Account: NetworkService
 Log File Path: NetworkService\Logs
 LocalSystem

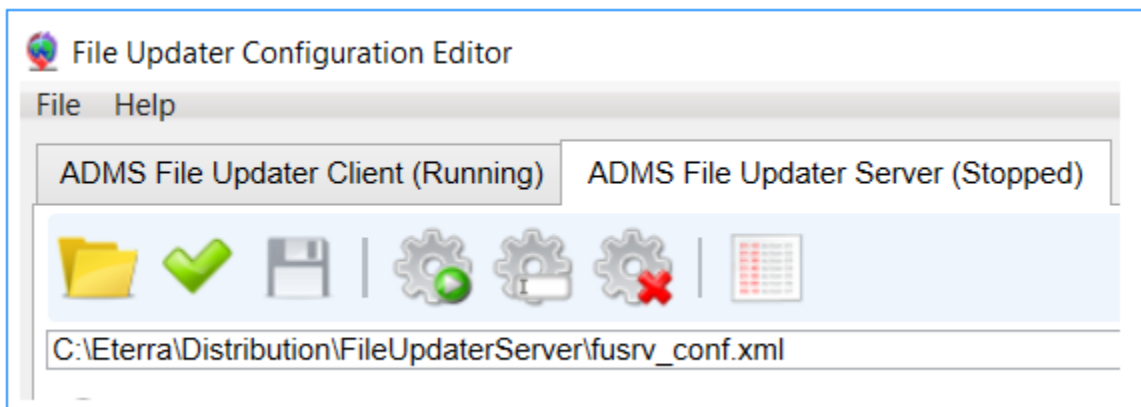
Max log file size (MB): 12

Updated Files Log: TEST.xml

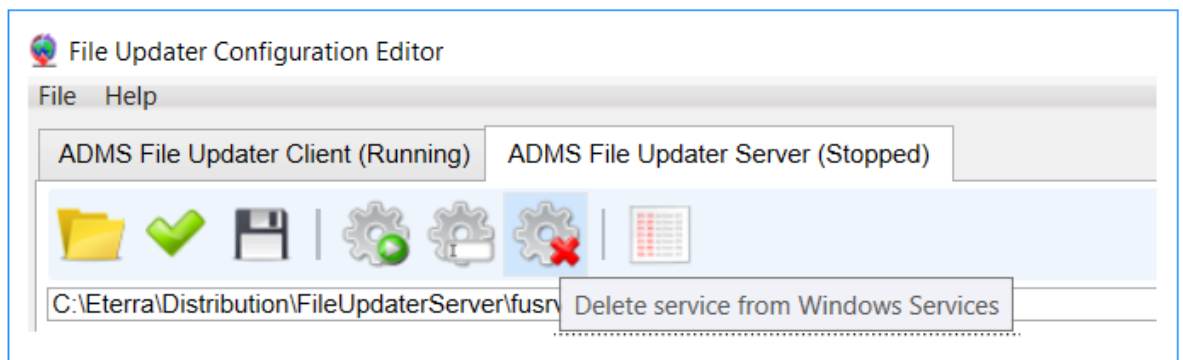
Retained log files count: -1

3. Click on the Save toolbar button.

The selected Account is saved in the service configuration file.



4. Click the Delete Service toolbar button.

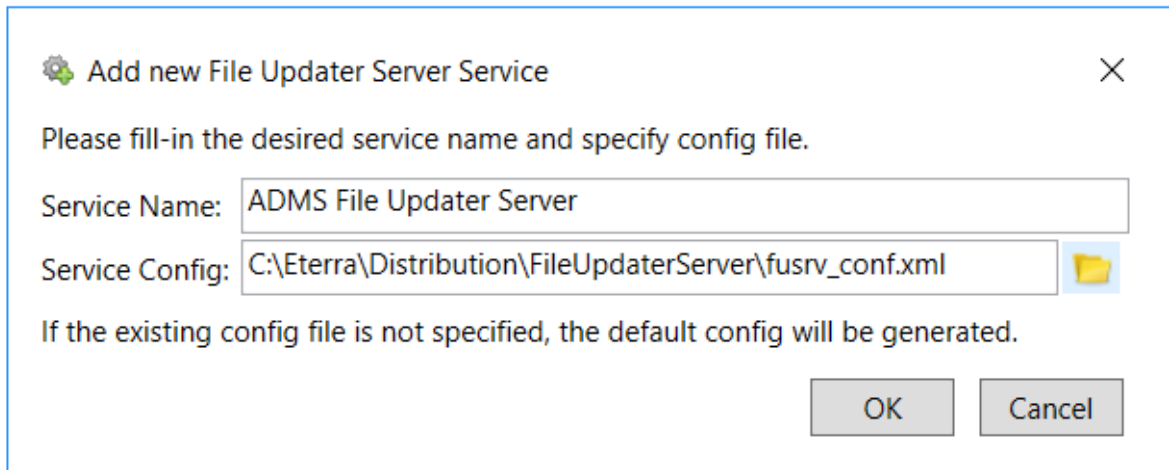


5. In Configuration Editor, select File > New > File Updater Service/Client.

The "Add a new File Updater Server/Client Service" dialog is displayed.

6. Set the Service Name.

7. Set the Service Config path.



 Add new File Updater Server Service ✕

Please fill-in the desired service name and specify config file.

Service Name:

Service Config: 

If the existing config file is not specified, the default config will be generated.

 Note

The Service Config path should point to the same configuration file which was updated when we saved the selected Account option.

8. Click the Start Service button.

5.4. File Updater Server Usage Notes

The File Updater server uses a local copy of the updated files log. A remote copy of the updated files log is stored on the Secured FTP (SFTP) server and is periodically synchronized by the File Updater server. During the synchronization, the remote copy is overwritten with a new version. This is a fully automatic process that does not need any user interaction.

If for any reason the remote copy of the updated files log is manually modified, all of the modifications are overwritten during the synchronization.

Copies of local files on the server are zipped locally in a temporary zip_temp folder. These files are then moved to the remote directories of the SFTP server. If the File Updater runs as a service, a prefix indicating the service instance name is appended to the temporary folder name.

Manual activities on remote copies of the files used for the synchronization (not only the updated files log) can lead to synchronization errors. This means that changes to a remote copy of this file do not affect the File Updater server. Remote copies are overwritten by a local copy of this file during the next update.

To force the upload of all files to the SFTP server (for example, in case remote files were damaged), delete the local copy of the updated files log (the local copy of this file has a prefix formed with the service instance name).

The File Updater server does not upload files until their copying process is completed.

A file copying process blocks the file from read access by Windows. The File Updater receives an

error when accessing this file and writes a message to the log file about the failed update attempt. On the next attempt, once the copying is completed, the File Updater server uploads the file. This prevents scenarios where the File Updater might upload a partially copied file.

When using several instances of the File Updater server with the same SFTP server, ensure that the UpdatedFilesLog value is different for each instance. Otherwise, the UpdatedFilesLog file of the first File Updater server instance will be overwritten by another instance's file. As a result, the File Updater client configured to work with the first instance of the File Updater server will be unable to perform an update.

5.5. File Updater Client Usage Notes

The File Updater client performs a CRC check on its local directories and compares this to what is stored in the UpdatedFilesLog to determine which files need to be synchronized. If for any reason some of the folders defined in the configuration files do not need to be synchronized, the corresponding nodes in the configuration files should not be specified. Only folders specified in the configuration files are synchronized, and there are no predefined or hard-coded folders for synchronization. Anything unspecified in the configuration file folders is mentioned in the warning messages in the synchronization log.

Note

The subfolders of the folders specified in the configuration files are not synchronized.

5.6. The Update Now Feature

To perform an update operation without waiting for the update timeout, use the following shortcut:

Start > Programs > Eterra > Distribution > File Updater > File Updater Client Update Now

A log message is written saying that the Update Now feature was called.

To use this function, an instance of the File Updater client should be running.

5.7. Detecting File Collisions

A "file collision" is an attempt to synchronize different files with the same name from different remote folders into the same local folder.

For example, you might need to synchronize all model files from many remote folders into one local folder. If two different remote folders have two different model files with the same name, File Updater synchronization attempts may be repeated. For example, the File Updater checks CRC against the file in the first remote folder. If the updater finds that the file folder is outdated, the File Updater updates it and checks it against CRC of the file from another folder. If that folder appears to be different, this process is repeated.

If directory entries are defined as dependent folders (entries with the same Time Delay value), File Updater can detect a file collision situation and report it via the log file.

5.8. Synchronizing Deleted Files

The File Updater server automatically deletes files in remote folders if the same files were deleted in a File Updater server local folder.

After the files are deleted by the File Updater server, the corresponding files are automatically deleted in the File Updater client's local folders during synchronization (if the directory entries define synchronization of those files).

Note

During synchronization of deleted files on the File Updater server, error messages may be printed in the File Updater client's log file. There is a short time period between when the File Updater server deletes remote files and when it updates the log file of updated files. Thus, if the File Updater client attempts to synchronize those files in that period, it detects the missing files in the remote folder. After the File Updater server ends synchronization, the File Updater client is able to delete the corresponding local files and successfully perform synchronization on its side.

5.9. Validation

The File Updater server and client have built-in configuration validation functions. Validation checks the File Updater server/client configuration file for common errors such as:

- Incorrect FTP/SFTP settings: host, login/password
- Non-existing remote/local folders
- Incorrect configuration paths

Validation can be performed from the Configuration Editor or using the command prompt. The following table classifies validation issues.

Table 1. Validation Issues Classification

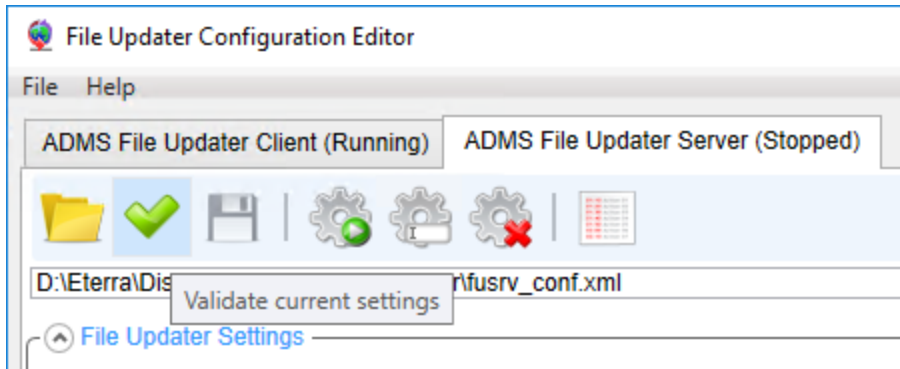
Validation	Applicable to	Severity
Configuration file cannot be correctly deserialized	Server/Client	Error
Local Log file path does not exist or is not available for read/write	Server/Client	Error
sLogFileType does not contain valid value	Server/Client	Error
sFileTransferProtocol field does not contain correct value	Server/Client	Error
Corresponding Protocol section does not contain all required values	Server/Client	Error
Server is not reachable via specified address, port, username and password	Server (FTP)/ Client (FTP)	Error

Validation	Applicable to	Severity
SecureFX is not installed or not available via sTool path	Server (SFTP)/ Client (SFTP)	Error
SecureFX's config folder does not exist (sToolConfigDir)	Server (SFTP)/ Client (SFTP)	Error
SFTP server (VShell) is not reachable	Server (SFTP)/ Client (SFTP)	Error
Remote directory is not accessible for read	Client	Error
Local directory does not exist or is not available for read/write	Client	Error
Remote directory is not defined or is not accessible for read/write	Server	Error
Time Delay value is incorrect	Server/Client	Error
File Updater Client failed to init. Please check configuration file. Error while initializing cryptography: The certificate signer [CERTIFICATE NAME] was not found.Error number [2148081668] System message [Cannot find object or property].	Server/Client	Error
Error while initializing cryptography: The certicate [CERTIFICATE NAME] can only be used after DATE TIME.	Server/Client	Error
Error while initializing cryptography: The certicate [CERTIFICATE NAME] has expired on DATE TIME.	Server/Client	Error
Could not sign file	Server	Error
File was not signed successfully	Server	Error
Error FILE_NAME file was not signed successfully. System Error: Could not sign file FILE_NAME. Error message: The certicate [CERTIFICATE NAME] has expired on DATE TIME	Server	Error
Could not verify the signature	Client	Error
Could not allocate memory to verify the signature	Client	Error
Could not calculate the size of the message to verify, signature verification failed	Client	Error
Check for Signature was skipped due to error on one of previous stages	Client	Error
Signature verification failed	Client	Error

5.9.1. Validating Using the Configuration Editor

To invoke validation from the Configuration Editor:

1. Click the green checkmark button.



2. Validation errors are highlighted and details are displayed in the Validation Log section.

Note

If you try to save a configuration file that has errors, the Configuration Editor displays a warning.

5.9.2. Validating Using the Command Line

To invoke validation from command line:

1. Run the File Updater server/client application with the "/validate" key.
2. Validation results are output to the console and can be redirected to a file using standard console redirection operators.

5.9.3. Validation Log File Structure

The first line is in the format:

%application_name | %config_file validation | %timestamp | %validation_type

where:

%application_name is "File Updater Client" or "File Updater Server".

%timestamp is date/time.

%validation_type is "local parameters validation", "remote parameters validation", or "full validation".

The lines that follow are validation messages in the format:

[%message_type] %parameter_name: %issue_description

where:

%message_type is "Error", "Warning", or "Info".

The last line (result line) is in the format:

[RESULT] %result

where:

%result is:

- PASSED
- PASSED (%N WARNINGS)
- FAILED (%N ERRORS)

%N is a count of errors/warnings.

5.10. Logging

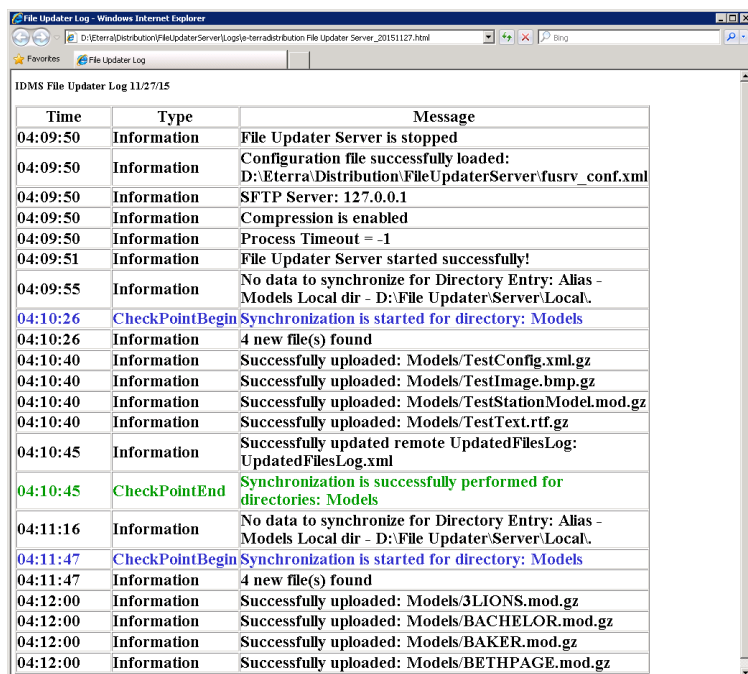
5.10.1. Log Files

The File Updater application logs are HTML files. The log files can be opened by any web browser.

The folder where log files are created is set in the configuration file.

Logs are written on a daily basis (one separate file each day). A log file name has the format ServiceName_YYYYMMDD.html.

"ServiceName" is the name of the File Updater service instance. The name indicates a service instance for log files, which is especially useful for configurations with several instances of the File Updater client (or server).



Time	Type	Message
04:09:50	Information	File Updater Server is stopped
04:09:50	Information	Configuration file successfully loaded: D:\Eterra\Distribution\FileUpdaterServer\fusrv_conf.xml
04:09:50	Information	SFTP Server: 127.0.0.1
04:09:50	Information	Compression is enabled
04:09:50	Information	Process Timeout = -1
04:09:51	Information	File Updater Server started successfully!
04:09:55	Information	No data to synchronize for Directory Entry: Alias - Models Local dir - D:\File Updater\Server\Local\.
04:10:26	CheckPointBegin	Synchronization is started for directory: Models
04:10:26	Information	4 new file(s) found
04:10:40	Information	Successfully uploaded: Models/TestConfig.xml.gz
04:10:40	Information	Successfully uploaded: Models/TestImage.bmp.gz
04:10:40	Information	Successfully uploaded: Models/TestStationModel.mod.gz
04:10:40	Information	Successfully uploaded: Models/TestText.rtf.gz
04:10:45	Information	Successfully updated remote UpdatedFilesLog: UpdatedFilesLog.xml
04:10:45	CheckPointEnd	Synchronization is successfully performed for directories: Models
04:11:16	Information	No data to synchronize for Directory Entry: Alias - Models Local dir - D:\File Updater\Server\Local\.
04:11:47	CheckPointBegin	Synchronization is started for directory: Models
04:11:47	Information	4 new file(s) found
04:12:00	Information	Successfully uploaded: Models/3LIONS.mod.gz
04:12:00	Information	Successfully uploaded: Models/BACHELOR.mod.gz
04:12:00	Information	Successfully uploaded: Models/BAKER.mod.gz
04:12:00	Information	Successfully uploaded: Models/BETHPAGE.mod.gz

Figure 14. File Updater HTML-Based Log File

5.10.2. Log Message Descriptions

A single log message is built from the following parts:

Time: The time when the message was sent to the logger. It is shown according to localization.

Type: There are four types of log message:

- Information: Information, status reporting messages
- Warning: Events that need attention
- Error: For failure reporting
- CheckPointStart, CheckPointEnd: Messages that indicate the start time and end time of the iteration. These messages structure log output and make it easy to read. CheckPointEnd indicates the successful execution of a section marked with CheckPointStart.

Message: The text of the message. It should be brief and informative. The message is intended to describe what happened and the reason for it, if possible (for example, writing error code).

Implemented Logging in File Updater

Information (including CheckPointStart, CheckPointEnd):

- Start and stop the application execution
- Process status (amount of outdated checked files, updated files, etc.)
- Begin, end of synchronization iteration for each folder set (File Updater client related)
- Update Now request received (File Updater client related)

Warning:

- SFTP server host changed (switch from primary to secondary and vice versa)
- Synchronization for specified folder set is retried

Error:

- Failure to create temporary folder.
- Failure to serialize generated settings.
- Failure to deserialize settings.
- Failure to deserialize updated files log.
- Failure to decompress downloaded file.
- Failure to perform file operation (copy, rename, delete).
- Failure to download or upload the file from SFTP. The error code returned from the SFTP client tool is added to the message.
- Failure to load digital certificate from local machine store while loading application. The error code returned from the cryptography system is included in the message.
- Failure to digitally sign files using local machine certificate. The error code returned from the cryptography system is included in the message.
- Failure to verify digitally signed files using local machine certificate. The error code returned from the cryptography system is included in the message.
- Any application exception. If it was not handled by the above messages, the system description (from the exception object) is shown.

CheckPointStart and CheckPointEnd messages contain log information with respect to the folder set that they refer to. Folder sets are defined in the File Updater client configuration file. There can be many folder sets in a specific configuration file. The folder set name is mentioned in the CheckPointStart message.

Whenever the File Updater client is running, it informs the user about any errors that have occurred. Some error messages do not refer to folder synchronization matters. If the File Updater client generates any errors before beginning folder synchronization, it writes all error messages to the log file or console.

In some cases during folder synchronization, error messages can be written to the console. This happens because of errors related to configuration file processing. For example, the configuration file may have undefined or corrupted log file settings. If log file settings are undefined, the File Updater client writes all errors to the console.

** Note**

If the log type (at the present time, only the HTML format is supported) is not specified correctly in the configuration file, all of the messages are written to the console. Therefore, if the File Updater application is started as a service, there is no way to read log messages.

5.10.3. System Log Usage

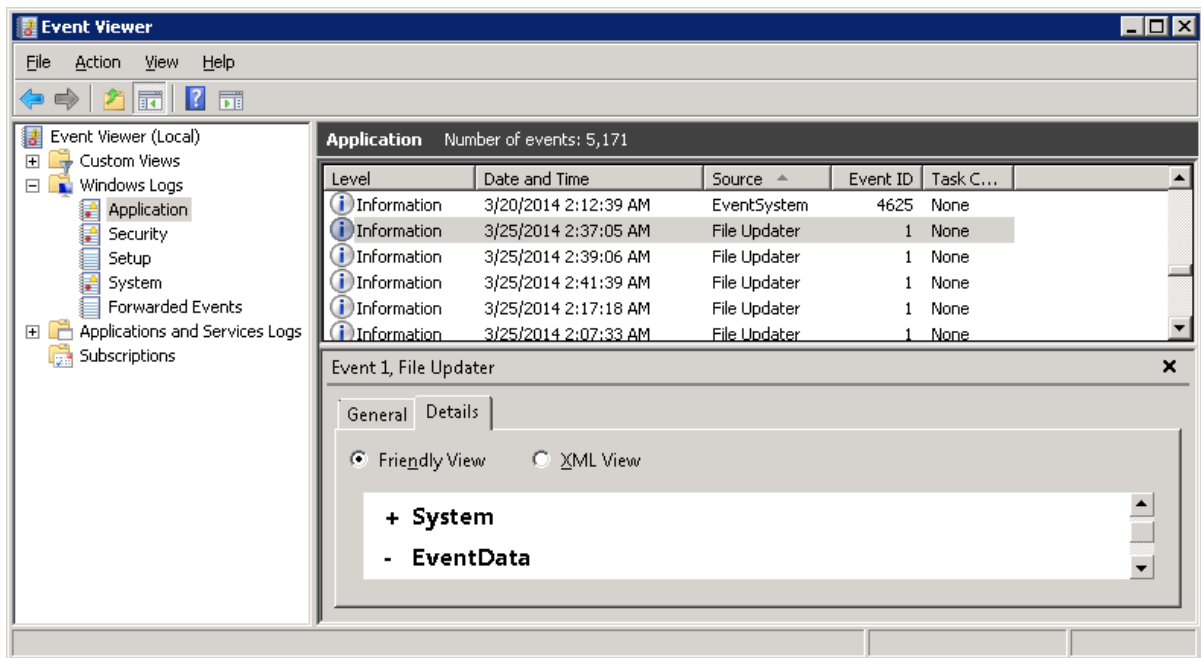
If an incorrect log file type is specified in the configuration file, the File Updater writes log messages into the system log.

The system log is platform-specific. It can be found in the locations listed below.

Windows:

1. Open Control Panel > Administrative Tools > Event Viewer.
2. In the left pane, select Windows Logs > Application.
3. Find and select the latest entry from the File Updater application (the Source column contains "File Updater").
4. In the Event pane, select the Details tab.

The log message text is shown in the EventData section.



Note that reading a system log file usually requires root access.

5.11. File Updater Command-Line Operations

Command line parameters are the same for both the File Updater server and the File Updater client. Replace *FileUpdaterExecutable* used below with:

- **File Updater Server:** IDMS_FileUpdaterServer.exe
- **File Updater Client:** IDMS_FileUpdaterClient.exe

The normal start-up command for the File Updater, as used for the Windows shortcut, is shown

below:

ExecutableLocation\FileUpdaterExecutable /c "configfile"

The File Updater can also be run directly from the command line.

The syntax for command-line operations is as follows:

- *ExecutableLocation\FileUpdaterExecutable [Autostart] /i "servicename" /c "configlocation\configfile"*

This command creates a service with the specified service name using the specified configuration file and executable location.

Note

Parameter values that are separated by spaces (service names, etc.) must be enclosed in quotes to be recognized as a single value.

- *FileUpdaterExecutable /u "servicename"*

Removes the specified service.

- *FileUpdaterExecutable /u all*

Removes all instances of the File Updater Server or Client service (depends on with which executable it is called).

- *FileUpdaterExecutable /l*

Prints a list of all instances of the File Updater Server or Client service (depends on with which executable the File Updater Server or Client service is called).

- *FileUpdaterExecutable /h*

Lists console-supported command-line arguments.

- *FileUpdaterExecutable /g "configfile"*

Generates the File Updater empty configuration file with given name.

- *FileUpdaterExecutable /c "configfile"*

Starts the File Updater using the specified configuration file.

- *FileUpdaterExecutable /v "configfile"*

Performs the configuration file validation. Validates the configuration file values, checks the connection with the FTP/SFTP server, and ensures that remote directories exist. For more details, see [Validation](#).

- *FileUpdaterExecutable /SETPASSWORD "configfile" password_string*

Sets and encodes the FTP password (sPassword field) for security.

6. Troubleshooting

If you are using an SFTP connection, the SSH connection validation procedure from [SFTP Installation and Configuration](#) may help troubleshoot most issues with SFTP.

6.1. Error Codes

When an error occurs during file upload or download, the File Updater Log is updated with a corresponding error code. Error codes are helpful when you investigate the reason for file transfer failures.

The following table contains the FTP protocol error code descriptions.

Table 2. FTP Error Codes Descriptions

Code	Description
100 Codes	The requested action is being taken. Expect a reply before proceeding with a new command.
110	Restart marker reply.
120	Service ready in (n) minutes.
125	Data connection already open, transfer starting.
150	File status okay, about to open data connection.
200 Codes	The requested action has been successfully completed.
200	Command okay.
202	Command not implemented.
211	System status or system help reply.
212	Directory status.
213	File status.
214	Help message.
215	NAME system type ("NAME" is an official system name from the list in the Assigned Numbers document).
220	Service ready for new user.
221	Service closing control connection (logged out if appropriate).
225	Data connection open, no transfer in progress.
226	Closing data connection. Requested file action successful (file transfer, abort, etc.).
227	Entering passive mode.
230	User logged in, proceed.
250	Requested file action okay, completed.
257	"PATHNAME" created.
300 Codes	The command has been accepted, but the requested action is being held pending receipt of further information.

Code	Description
331	User name okay, need password.
332	Need account for login.
350	Requested file action pending further information.
400 Codes	The command was not accepted and the requested action did not take place. The error condition is temporary, however, and the action may be requested again.
421	Service not available, closing control connection. (May be a reply to any command if the service knows it must shut down.)
425	Cannot open data connection.
426	Connection closed, transfer aborted.
450	Requested file action not taken. File unavailable (for example, file busy).
451	Requested action aborted, local error in processing.
452	Requested action not taken. Insufficient storage space in system.
500 Codes	The command was not accepted and the requested action did not take place.
500	Syntax error, command unrecognized. This may include errors such as command line too long.
501	Syntax error in parameters or arguments.
502	Command not implemented.
503	Bad sequence of commands.
504	Command not implemented for that parameter.
530	User not logged in.
532	Need account for storing files.
550	Requested action not taken. File unavailable (for example, file not found, no access).
552	Requested file action aborted, storage allocation exceeded.
553	Requested action not taken. Illegal file name.

SFTP protocol (Vandyke SecureFX error codes):

According to Vandyke support, error code descriptions are not defined and error codes values may vary from version to version.

6.2. Running File Updater in Debug Mode

File Updater can be run in debug mode to generate a SecureFX command-line tool (sfxccl) log file that contains Secure FTP file transfer details. The log is recorded in the Log File Path directory and has the "sfxccl_" prefix.

For example:

- sfxccl_20210720.log, when running the File Updater in command line mode

Or:

- `sfxcl_ADMS File Updater Client_20210720.log`, when running the ADMS File Updater Client service.

To run File Updater in debug mode:

1. Open a command prompt.
2. Execute the following command:

```
setx /m FILE_UPDATER_DEBUG_MODE TRUE
```

This adds a system-wide `FILE_UPDATER_DEBUG_MODE` environment variable and sets the environment variable value to `TRUE` (case sensitive).

To run File Updater in normal mode:

1. Open a command prompt.
2. Execute the following command:

```
setx /m FILE_UPDATER_DEBUG_MODE FALSE
```

6.3. Log Files Are Not Being Created

Problem: The File Updater application is running. However, in the specified log folder (`sLogFilePath`), the log file was not created.

Solution 1: Check to see if the log type was properly set in the configuration file (at the present time, only the HTML format is supported).

Solution 2: Ensure that the File Updater application has rights to write and to create files within the directory specified as the log folder (`sLogFilePath`).

6.4. Upload or Download Errors Are Reported

Problem: The File Updater application reports an upload or a download error.

Explanations and Solutions: This can happen for the following reasons:

- Improper SFTP-related settings are specified in the configuration file.
 - Ensure that the specified public key file exists and is accessible for the SecureFX application. Also ensure that the public key is properly populated to the SFTP server by connecting to it via the SecureFX GUI application.
 - Ensure that the SFTP server IP address is correct.
 - Ensure that the SecureFX command-line tool is correctly specified.
 - Ensure that the user name (login) is correctly specified.
 -

Ensure that the remote folders are correctly specified.

- Environment problems
 - Ensure that the SecureFX application directory contains the Config subfolder.
 - If firewall software is running, ensure that the firewall does not block the outgoing connection with SecureFX.

6.5. "Can't resolve local folder for remote folder" Log Message

Problem: A retrieved log message reads "Can't resolve local folder for remote folder".

Explanation: This means that remote folder paths specified in the File Updater server configuration file do not correspond to paths specified in the File Updater client.

This happens because the remote folder is specified by the File Updater server in the updated files log file. Then the File Updater client retrieves this file and tries to bind this remote folder to the local folder according to the configuration file. If the remote folder path value specified in the updated files log is not equal to a value specified in the File Updater client configuration file, the application is unable to resolve which folder this remote folder corresponds to.

6.6. File Updater Fails if Files Have Non-English File Names

Problem: The File Updater application successfully processes files with English file names. However, it fails if files have non-English file names (Chinese, Russian, etc.).

Explanations and Solutions: There are two reasons this can be happening:

- The version of SecureFX being used does not correctly support Unicode file names. Install SecureFX 6.1 or later.
- The configuration parameter "sSFTPEncoding" is not set correctly. In this case, set "sSFTPEncoding" to UTF16.

6.7. New Log Records Are Not Appearing

Problem: The log file exists, but there are no new records in it.

Explanations and Solutions: There can be several reasons for this:

- Ensure that the File Updater application is running.
- Ensure that you have examined the correct file. Log files are created on a daily basis. If the date has changed since the last log file observation, new log messages are added in the new log file.

- Ensure that access rights to the folder where the log files should be placed have not been changed and that this folder has not been deleted.
- Ensure that there is no error in the configuration file regarding the path to the log files folder and the type of log file to use.

6.8. The File Updater Application Service Is Not Starting

Problem: The File Updater application service does not start.

Explanations and Solutions: In most cases, the reason is errors in the configuration file. Attempt to start the application from the command line. When starting from the command line, the File Updater application informs the user about configuration-related errors via the console.

If the File Updater application starts successfully from the command line but is unable to start as a service, reinstall the File Updater application.

For more information, see [File Updater Command-Line Operations](#).

6.9. Changed Data Files Are Not Restored in Future Updates

Problem: For the File Updater client, a data file in local folders has been changed (modified or deleted), but it has not been restored for the next update.

Explanations and Solutions: The File Updater client was not designed to perform monitoring of local data files. The File Updater client monitors remote files for changes, and it checks local files only when a change of remote files occurs. If local files were damaged, restart the File Updater client to restore them; otherwise, those files are restored automatically after the next change of remote folders.

6.10. "Download of UpdatedFilesLog failed. ErrorCode: -1" Log Message

Problem: The log file contains this message:

```
Download of UpdatedFilesLog failed. ErrorCode: -1
```

This problem appears in Windows.

Explanation and Solution: This message informs the user that the log file failed to run the `sfxcl.exe` file, which is a part of the VanDyke software used to connect to the SFTP server. Check the configuration file of the File Updater client, and verify the full path to the `sfxcl.exe` file that is defined during configuration.

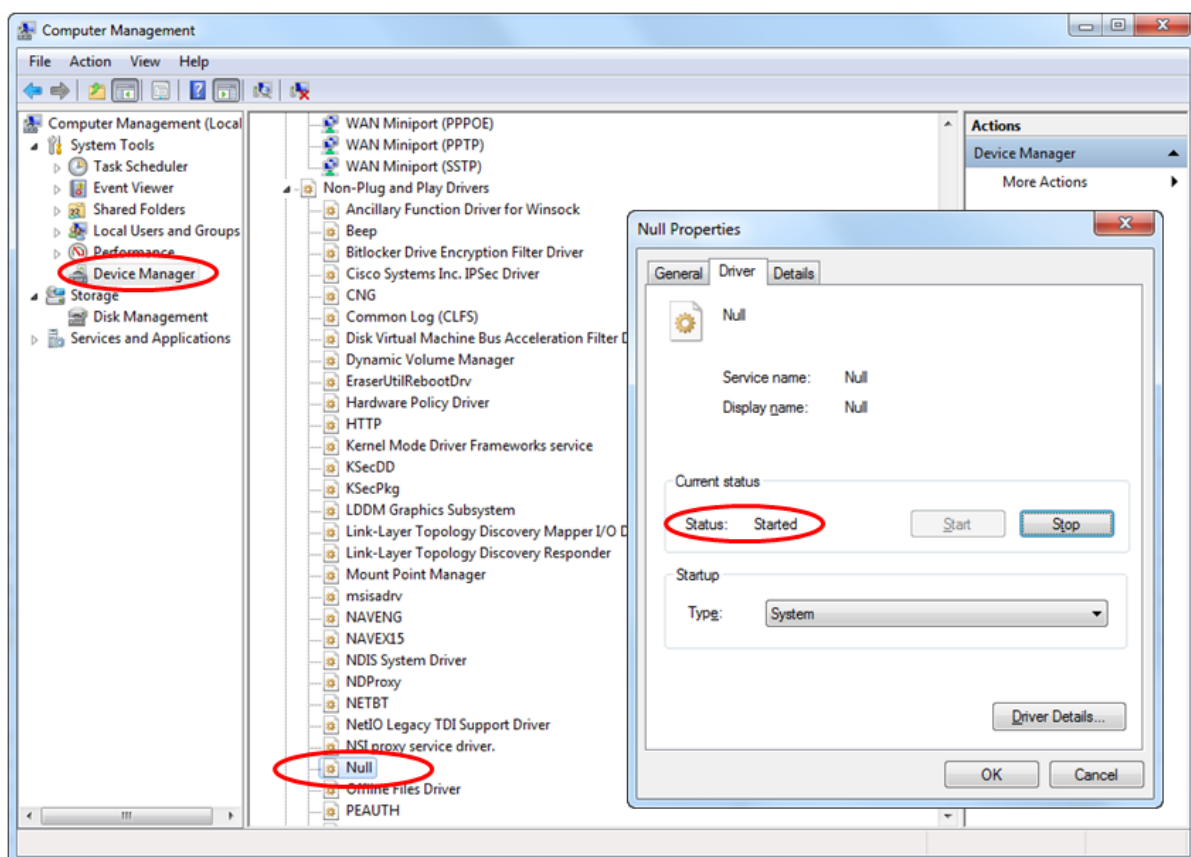
Problem: The log file contains this message:

```
Download of UpdatedFilesLog failed. ErrorCode - -1 : Error while downloading file:  
UpdatedFilesLog.xml: boost::process::detail::win32_start: CreateFile failed: The system cannot find  
the file specified. For more information check the Vandyke SecureFX/VShell logs.
```

Explanation and Solution: As the SecureFX redirects output to the Null device, this problem may appear if the Null device is not working properly, or its driver is not started.

To start the Null device driver:

1. Navigate to Start > Control Panel > Administrative Tools > Computer Management.
2. Select Device Manager.
3. On the View menu, select Show Hidden Devices.
4. Expand the Non Plug and Play Drivers tree.
5. Scroll to the Null selection, and select Properties from the context menu.
6. Enable the Null device and start its driver.



6.11. Uploads or Downloads Are Failing

Problem: The File Updater server/client has been failing to upload or download a file. At the same

time, SFTP settings do not show obvious mistakes.

Explanations and Solutions: The reason for this may be the use of a passphrase public key. To check the public key, open the SecureFX GUI application, and create a connection to the SFTP server using only public key authorization.

To solve the problem, regenerate the public key without the passphrase, and propagate it to the SFTP server according to the *Secure Shell (SSH) User Guide*. After propagating the key, modify the configuration file of the File Updater application.

6.12. Uploads Are Successful but Downloads Fail

Problem: The File Updater server reports that a file has been successfully uploaded. At the same time, the File Updater client is unable to download the given file, and the second letter in the file name is "s".

Explanation and Solution: The problem refers to the area of SecureFX responsibility. To solve the problem, install SecureFX 6.1.4 or later.

6.13. Files Are Not Uploaded or Downloaded After the Update Timeout

Problem: The File Updater server/client failed to upload or download a file that does not perform any actions after the next update timeout. Usually, the File Updater server/client is supposed to upload or download the file after the update timeout.

Explanation and Solution: The problem is the SFTP connection settings (passphrase authentication). If the public key uses passphrase authentication, the SFTP server waits for the passphrase. To solve the problem, reconfigure the SFTP connection so that a passphrase is not used, and propagate a new public key to the SFTP server.

6.14. Log File Reports Successful Operation But It Was Not Performed

The Log file contains a message that files were successfully uploaded, but the remote folders contain no files. A similar issue may occur when downloading files.

Problem: SecureFX failed to perform the SFTP operation but returned a successful error code.

Explanation and Solution: Verify that the path to the secure FTP tool configuration folder (sToolConfigDir) is defined correctly in File Updater's configuration file.

6.15. Unable to Connect to FTP on Local Host

Problem: If bSFTPserverIsLocal is set to 1, 127.0.0.1 is used as the FTP server IP address.

Explanation and Solution: In the IIS Manager FTP site identification, verify that the FTP site properties are defined as "All unassigned". If there is an IP address specified, you should specify this IP in the File Updater configuration file and set bSFTPserverIsLocal to 0.

6.16. Unable to Read Configuration File

Problem: The File Updater cannot read the configuration file. The following error occurs during the validation:

```
[Error] Unable to read configuration file. The configuration file structure might be broken.
```

Explanation and Solution: To solve this problem, verify the structure of the configuration file, and also check the versions of File Updater and the configuration file. This problem may occur due to a version mismatch.

6.17. Exception While Reading Configuration File, Reason: Stream Error

Problem: The File Updater cannot read the configuration file.

When starting the File Updater client manually using the following command:

```
IDMS_FileUpdaterClient.exe /c "<configuration file name>"
```

The following message prints in the console:

```
[Information] File Updater Client is starting...  
[Error] Exception while reading configuration file `fuclt_conf.xml`  
Reason: stream error
```

Explanation and Solution:

Possible reasons:

- The <count> node has the wrong value.
- The original XML structure is damaged.
- There is no <!DOCTYPE boost_serialization> tag.

Compare the structure of the configuration file used with the newly generated configuration file.

6.18. Manually Installing the File Updater Server/Client

The File Updater installation kit performs an automatic installation that does not require interaction with an end user (support engineer). However, these tasks can also be performed manually. In some cases, manual installation may be required.

Knowledge of the manual installation process can help in troubleshooting or when, for any reason, automatic installation cannot be performed.

6.18.1. Manual Installation Prerequisites

The support engineer must have successfully configured the secured data transfer between a File Updater server computer, ADMS server computers, and ADMS client computers.

The support engineer must have installed and configured the recommended secured data transfer utilities on the different nodes, according to the recommended GE standards.

6.18.2. Manually Installing the File Updater Server

To manually install the File Updater server:

1. Create the File Updater server/client configuration file (for example, `fusrv_conf.xml`).

To generate an empty configuration file, run the following command:

```
IDMS_FileUpdaterServer.exe /g fusrv_conf.xml
```

2. Go to the folder where the application binaries reside.
3. Run the following command:

```
<File Updater server executable dir>\IDMSFileUpdaterServer.exe /i "ADMS File Updater Server" /c "<config file>"
```

i Note

`<config file>` is the full configuration file name path. The entire "`<config file>`" parameter must be inside quotation marks.

i Note

Replace the path `<File Updater server executable dir>` with the path to the File Updater server executable file.

4. Create the following shortcuts:

- Start_FileUpdaterServer_Service

command: `NET START "ADMS File Updater Server"`

- Stop_FileUpdaterServer_Service

command: `NET STOP "ADMS File Updater Server"`

Note

The entire "<service name>" parameter must be inside quotation marks.

6.18.3. Manually Installing the File Updater Client

To install the File Updater server manually:

1. Create the File Updater client configuration file (for example, `fucflt_conf.xml`).

To generate an empty configuration file, run the following command:

```
IDMS_FileUpdaterClient.exe /g fucflt_conf.xml
```

A more-detailed description of configuring the File Updater client is specified in [Configuring the File Updater Server/Client](#).

Note

Replace the path <File Updater client executable dir> with the path to the File Updater client executable file.

2. Go to the folder with the application binaries.
3. Run the following command:

```
<File Updater client executable dir>\IDMSFileUpdaterClient.exe /i "ADMS File Updater Client" /c "<config file>"
```

Note

<config file> is the full configuration file name path. The entire "<config file>" parameter must be inside quotation marks.

Note

Replace the path <File Updater client executable dir> with the path to the File Updater client executable file.

4. Create the following shortcuts:

- Start_FileUpdaterClient_Service

command: `NET START "ADMS File Updater Client"`

- Stop_FileUpdaterClient_Service

command: `NET STOP "ADMS File Updater Client"`

- FileUpdaterClient_UpdateNow

command: "<File Updater client executable dir>\IDMS_FileUpdaterClient.exe" /UPDATE

Note

Replace the path <File Updater client executable dir> with the path to the File Updater client executable file.

6.19. Starting the File Updater Client in Debugging Mode

The File Updater client, when started, writes to a log file only after every significant step. Using the commands below, the client can be started in a debugging mode where the logging includes detailed information, i.e. information on current application state.

- To run the File Updater client as a console application, enter the following command at the command prompt:

```
D:\Eterra\Distribution\FileUpdaterClient\IDMS_FileUpdaterClient.exe /C
D:\Eterra\Distribution\FileUpdaterClient\fuclt_conf.xml /debug
```

- To run the File Updater client as a service, enter the following command at the command prompt:

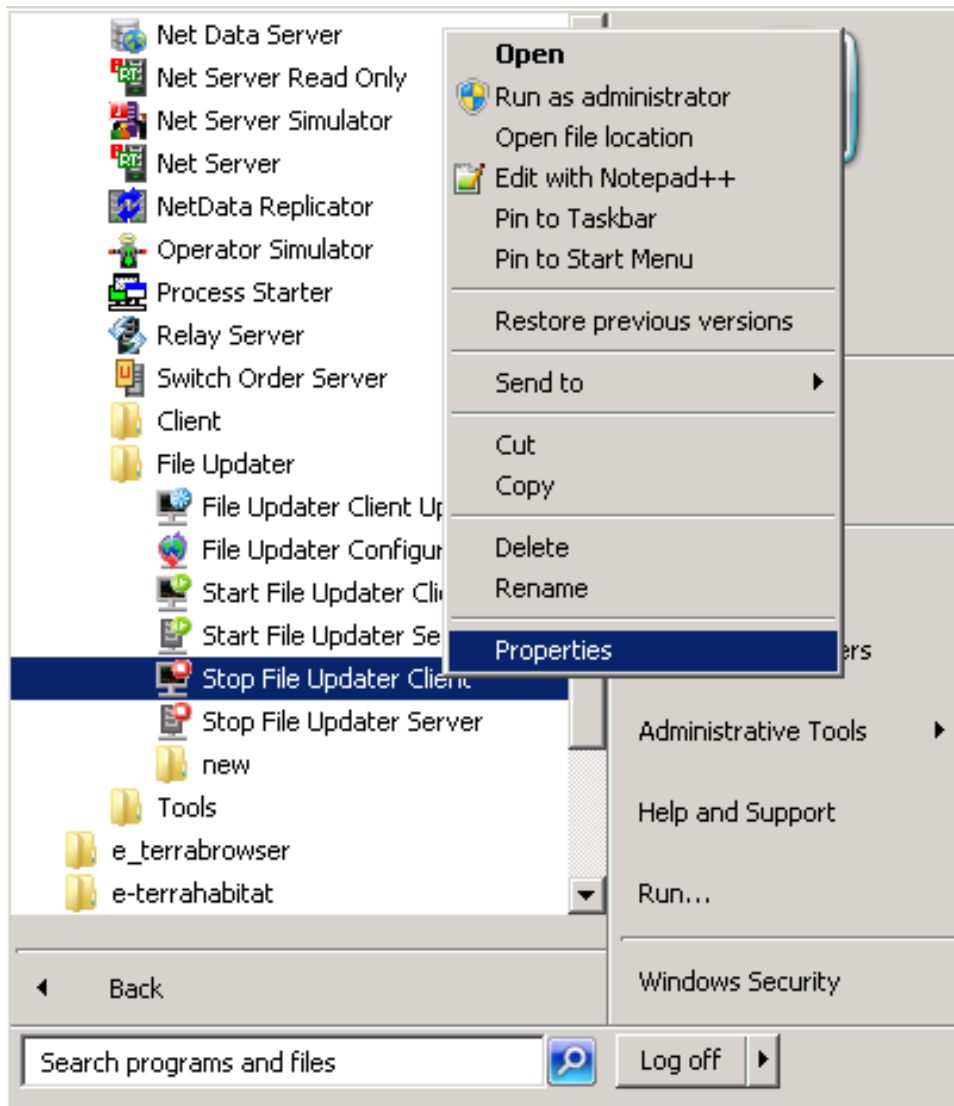
```
D:\Eterra\Distribution\FileUpdaterClient\IDMS_FileUpdaterClient.exe /c
"D:\Eterra\Distribution\FileUpdaterClient\fuclt_conf.xml" /s "ADMS File Updater Client" /debug
```

6.20. Setting Privileges for the File Updater Shortcut

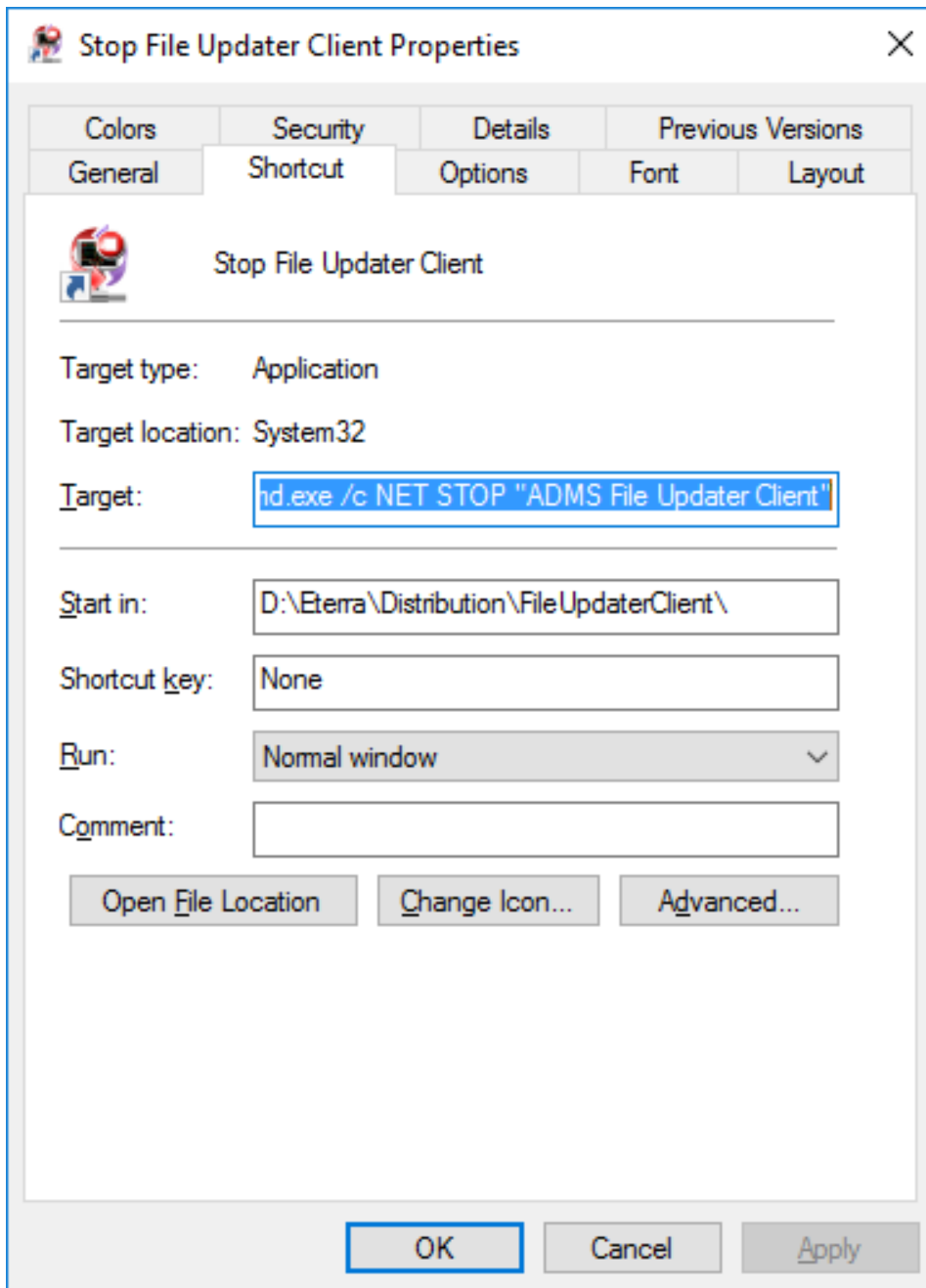
In Windows 7 and Windows Server 2008, to start/stop programs that run as Windows services requires administrator privileges. If a program starts as a Windows service, it needs to be granted administrative rights.

To set administrator privileges for the File Updater server shortcut:

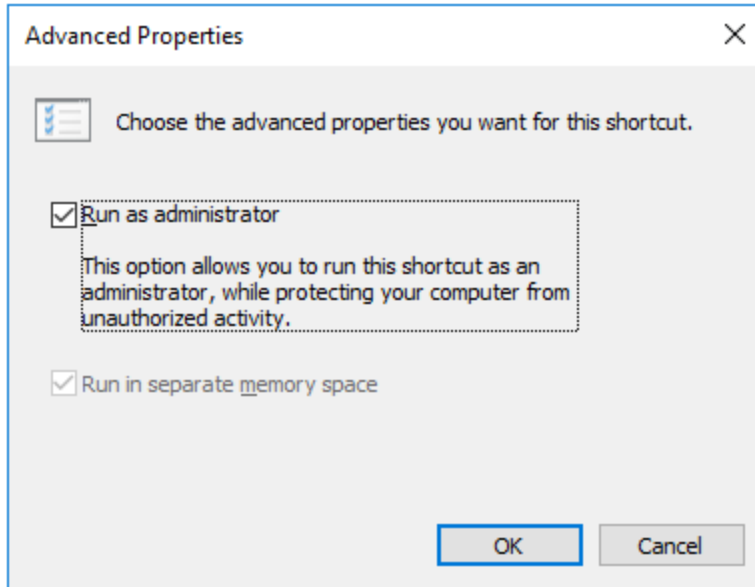
1. Open the Windows Start menu and locate the shortcut.
2. Right-click the shortcut and select Properties.



3. In the Properties dialog box, select the Shortcut tab.



4. Click the Advanced button.
5. On the Advanced Properties screen, select Run as Administrator.



6. Click OK.

This closes the Advanced Properties window and navigates to the parent window (the File Updater Server Shortcut Properties window).

7. In the Shortcut Properties window, click OK.

6.21. Error Message in Log File: Error while uploading: <file_name> ErrorCode: 255

Ensure that the sToolConfigDir configuration file parameter contains the correct path to the configuration directory of VanDyke's SSH command-line tool. More details are in [Configuring the File Updater Server/Client](#) (see the description of the sToolConfigDir node).

6.22. Error Message in Log File: Error while uploading: <file_name> ErrorCode - -516095934

Explanation: This error was observed when using VanDyke SecureFX v 8.0.3. A public key that was generated by the VanDyke SecureFX application using the Standard Public Key and VanDyke Private Key Format option was not correctly applied by the File Updater.

Solution: Use the OpenSSH Key Format (Legacy) option to generate a public key.

6.23. File Updater Client does not Switch to the Secondary SFTP Server when the Primary Server is Unavailable

Explanation: The Server Unavailability Error Code value in the SecureFTP configuration does not correspond to the server unavailability error code of the applied version of SecureFX.

The Server Unavailability Error Code integer value is used by the "server switch" functionality. This parameter is used only in Windows with SFTP.

The SecureFX utility has a specific error code for cases when the SFTP server is unavailable (VShell or the server host itself). Usually, this error code value matches the Server Unavailability Error Code parameter in the generated configuration file. When the error code is returned from SecureFX, it tells File Updater about the SFTP server unavailability.

Solution: It is recommended that you use the default value of the Server Unavailability Error Code parameter provided in the generated (new) configuration file. If the default value does not work as expected, locate the expected Error Code value using the steps below:

1. Configure the File Updater to use the unavailable SFTP server. For more information about configuring the File Updater, refer to [Server Switch Feature](#).
2. Initiate the file transfer.
3. In the log file, locate a failed operation message entry with the expected error code value. For example:

```
"Connection test to the server Host2 failed, error code: -519765975 while server unavailability error code is: -100"
```

4. When you identify the expected error code, enter the value in the Server Unavailability Error Code field of the SecureFTP configuration.

6.24. File Updater stops due to invalid Digital Certificates.

- Explanation: Digital certificate invalid errors are reported then service is stopped.
- Solution: If Digital Signature is enabled, File Updater will check for certificate availability and validity (start and expiration dates are valid) during startup and every time a file will be digitally signed.

6.25. File Updater is not able to digitally sign files or verify digital signatures in files

- Explanation: Digital signature errors are reported when the application is not able to open digital certificate installed in local machine to digitally sign files (server side) and later verify signatures (client side).
- Solution: To enable the digital signature functionality, it is required to have a valid certificate obtained through either source:
 - Self-signed certificate.
 - Certificate provided by a Certification Authority (CA). It is recommended to install the certificate at Local Machine level in the Trusted Root Certificates Authorities. You can verify the existence of the certificate using the Windows Certificate Management Tool. Once installed, the value to configure in the Certificate Store Name is "root". The Certificate Name is the one specified in your certificate. The certificate name is usually displayed in the first column in the list of certificates displayed by certmgr.exe.

6.26. Important Notes

- If the File Updater application exits because of any errors occurring (for example, configuration file corrupted), the corresponding service is still displayed in Control Panel > Administrative Tools > Services with a "running" state.

To ensure that the File Updater application is running, check the Task Manager processes list. That list should include either the IDMS_FileUpdaterServer process or the IDMS_FileUpdaterClient process (depending on which application you are interested in).

It is recommended that you look at the log file if the File Updater application exits. However, there may be situations in which the File Updater application is unable to write to the log file (application is terminated by the operating system, access to the file system is prohibited, etc.).

- The File Updater application does not automatically reload the configuration file. If the configuration file was changed, the File Updater application should be manually restarted.
- There are known issues around VanDyke SecureFX that result in the File Updater application being unable to inform the user about errors:
 - The SecureFX Config folder is placed in a different directory from the SecureFX executables. This situation triggers the following:
 - The File Updater server informs the user that all files were successfully uploaded to the SFTP server, when this is not the case.
 - The File Updater client informs the user that all files were successfully downloaded, but decompression failed.

Since all needed information regarding the log can be retrieved by the File Updater application

only after successful loading of the configuration file, all log messages regarding errors and events that occur before this moment are output to the console. Also, if the File Updater application is unable to write to the log file, or if its type specified in the configuration file is incorrect, the File Updater application writes log messages to the console Windows.

It should be taken into consideration that, if the File Updater application is running as a service, log messages put in the console are not visible.

- The File Updater server does not monitor remote files. This means that deleting or changing remote files does not cause the File Updater server to upload the local file with the same name (this happens only if the file was modified locally).
- Retry-related parameters in the configuration file are related to each connection to the SFTP server, not to the File Updater application operation. Therefore, the number of retries means how many attempts are made to establish a connection with the remote SFTP server, not the number of File Updater application update operation attempts.
- Most problems occur because of incorrect configuration files and improper settings of SSH third-party software. Therefore, when File Updater-related problems occur, it is recommended that users first check the File Updater application configuration file and the SSH client software.
- It is assumed that support engineers do not place files with the same name in different folders (the term "file name" means the pair of a file name and a file extension). Files of the same type must be stored in specific folders. If any folders contain files with the same name, it leads to synchronization errors.
- When you use the FTP protocol for file transfer, refer to the FTP error codes in [Error Codes](#).